School of Informatics & Cybersecurity - Assessment Submission Coversheet

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Module Information

Module Title:	Security Administration
Module Code:	COMP H2703

Assessment Information

Assessment Title:	Project

Declaration

☒ I confirm that I have read and understood the University's Academic Integrity Policy.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/academic-quality-assurance-and-enhancement/academic-integrity/

☒ I have completed the TU Dublin AI Driving License theory test.

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☒ I declare that this assessment submission is my own work.

☒ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

Student signature:	Roisin Aylward
Date:	01/12/2025

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Summary

The report documents the design and implementation of a cloud infrastructure for the software development company, TechStart Solutions' web application. Microsoft Azure was used as the cloud platform to meet the company security requirements that consisted of network segmentation, least-privilege access control, strong authentication, encryption in transit and at rest, system hardening and system monitoring, and a logging, backup, audit trail. Ubuntu Linux was used as the Virtual Machine to host the web application using Apache, mod_wsgi, and Flask.

Architecture Overview

A resource group named TechStart-Solutions was created to contain and manage the different resources used together. The resources were mainly located in Norway East to reduce latency and to keep costs low. There is a single Azure Virtual Network with three subnets to isolate the Ubuntu web server VM, the SQL database, and the storage.

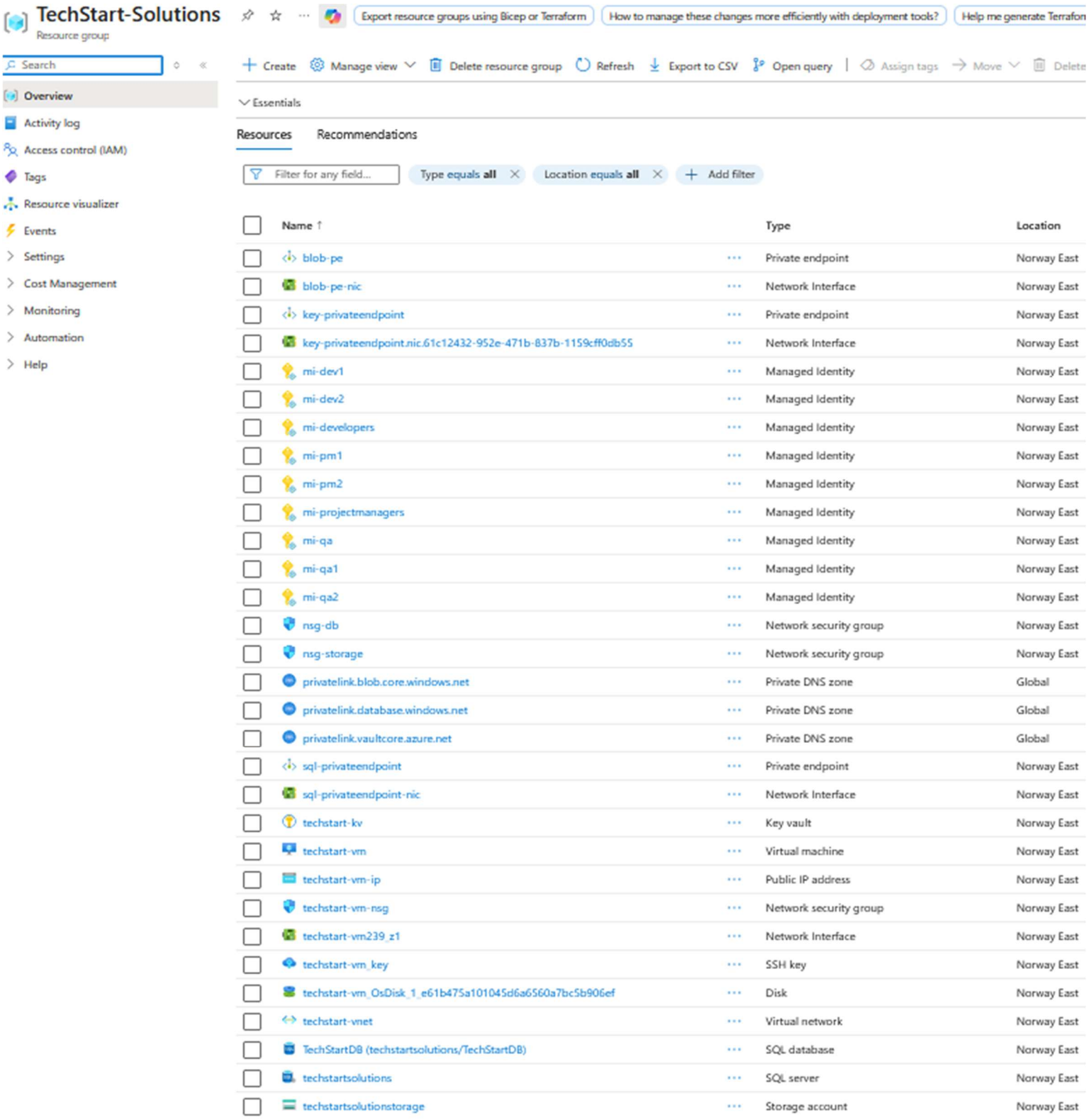


Figure 1 above: All of the resources in TechStart-Solutions resource group.

techstart-vnet
VIRTUAL NETWORK




subnet-db
SUBNET
sql-privateendpoint-nic
key-privat…59cff0db55
key-privateendpoint
privateEn…ea750a6ae
nsg-db
sql-privateendpoint
privateEn…66a2a154a



Figure 2 above: Network topology diagram of Virtual Network containing subnets, NSGs, private endpoints, and the Virtual Machine.

Security Implementation

Network Segmentation and Access Control

Configuration

A Virtual Network called techstart-vnet was given the address space 10.0.0.0/16 for three subnets. subnet-web was divided into 10.0.1.0/24 and deployed the Ubuntu VM (techstart-vm) which has 10.0.1.4 as its address. techstart-vm has a network interface (techstart-vm239_z1) that is attached to a public IP address (techstart-vm-ip) of 131.163.89.9, and a Network Security Group (techstart-vm-nsg) which has inbound and outbound rules following the principle of least privilege. The inbound rules in the order of priority allow SSH (TCP port 22) only from my public IP (and can be changed on the Azure portal when I need to remotely access techstart-vm from a different network) to enable configuration on techstart-vm, and allow HTTP (TCP port 80) and HTTPS (TCP port 443) from any source for requests to access the web application login page. The inbound rules were also preconfigured to allow any traffic originating from the rest of techstart-vnet to enable communication with the storage and SQL database, and with a default deny rule that blocks all other traffic from sources elsewhere to prevent unauthorised access. The outbound rules were also preconfigured in the order of priority to allow any traffic destined to the rest of techstart-vnet to also enable communication, allow any traffic going to the internet to respond to requests for the web application, and deny any other traffic going anywhere else.

subnet-storage was divided into 10.0.3.0/24 and deployed a storage account (techstartsolutionstorage) with containers that store blobs for the web application. subnet-storage has a Network Security Group (nsg-storage) that follows the principle of least privilege. The inbound rules in the order of priority allows HTTPS from the subnet-web to the subnet-storage and does not get forwarded, and denies any other traffic from any sources including other subnets on techstart-vnet. There was a default deny rule already preconfigured, but it had a lower priority than another preconfigured rule that would allow any traffic originating from the rest of techstart-vnet, which could not be deleted or changed and therefore required a duplicate default deny rule that had a higher priority.

subnet-db was divided into 10.0.2.0/24 and deployed an SQL server (techstartsolutions) which has an SQL Database (TechStartDB). Public network access is disabled on techstartsolutions SQL server, instead its network interface (sql-privateendpoint-nic) has a private endpoint (sql-privateendpoint) with 10.0.2.4 as its address. subnet-db has a Network Security Group (nsg-db) with an inbound rule that allow Microsoft SQL Server (TCP port 1433) from techstart-vm to sql-privateendpoint. Like nsg-storage, a custom default deny rule with a higher priority has to be configured as the preconfigured default deny rule had a lower priority than the preconfigured rule that a would allow any traffic originating from the rest of techstart-vnet.

Screenshots

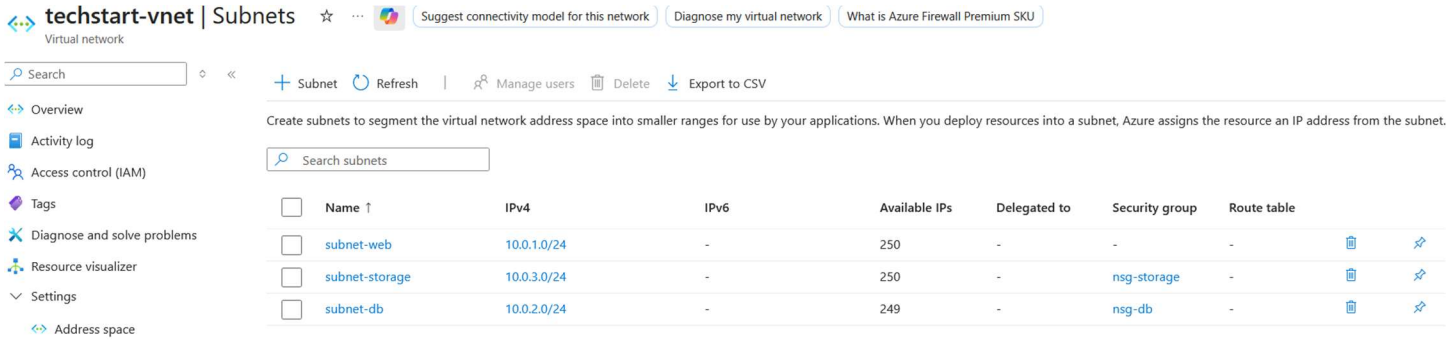


Figure 3 above: techstart-vnet's three subnets: subnet-web, subnet-storage, and subnet-db.

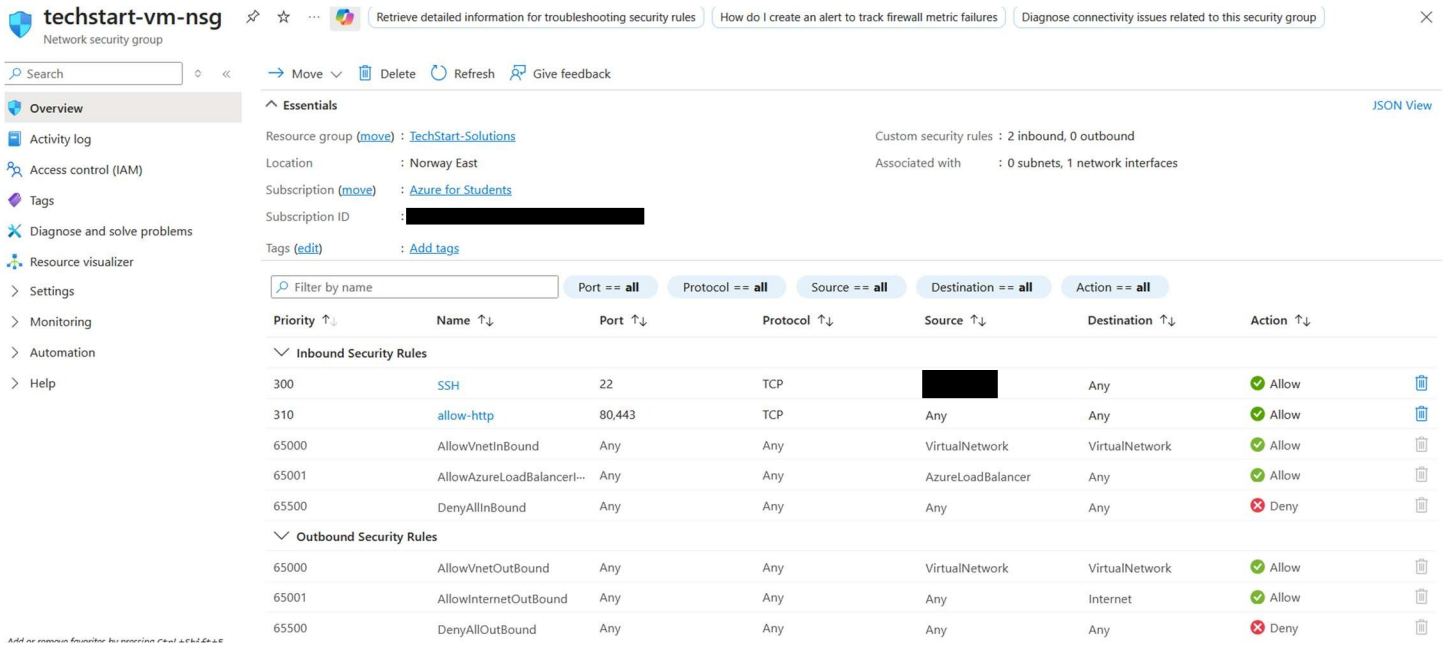


Figure 4 above: techstart-vm-nsg's security rules where inbound only HTTP/HTTPS for web traffic & SSH from my public IP allowed.

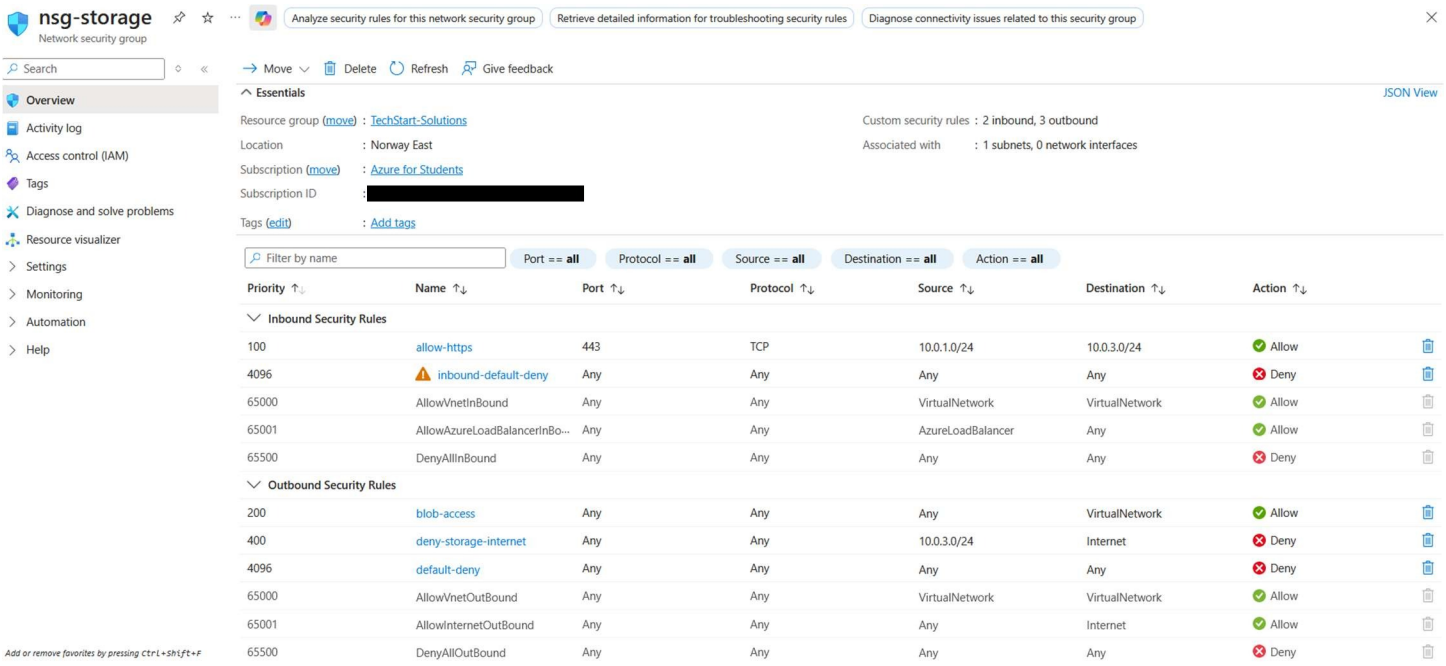


Figure 5 above: nsg-storage's security rules which allows only incoming HTTPS traffic and outgoing blob and internet access.

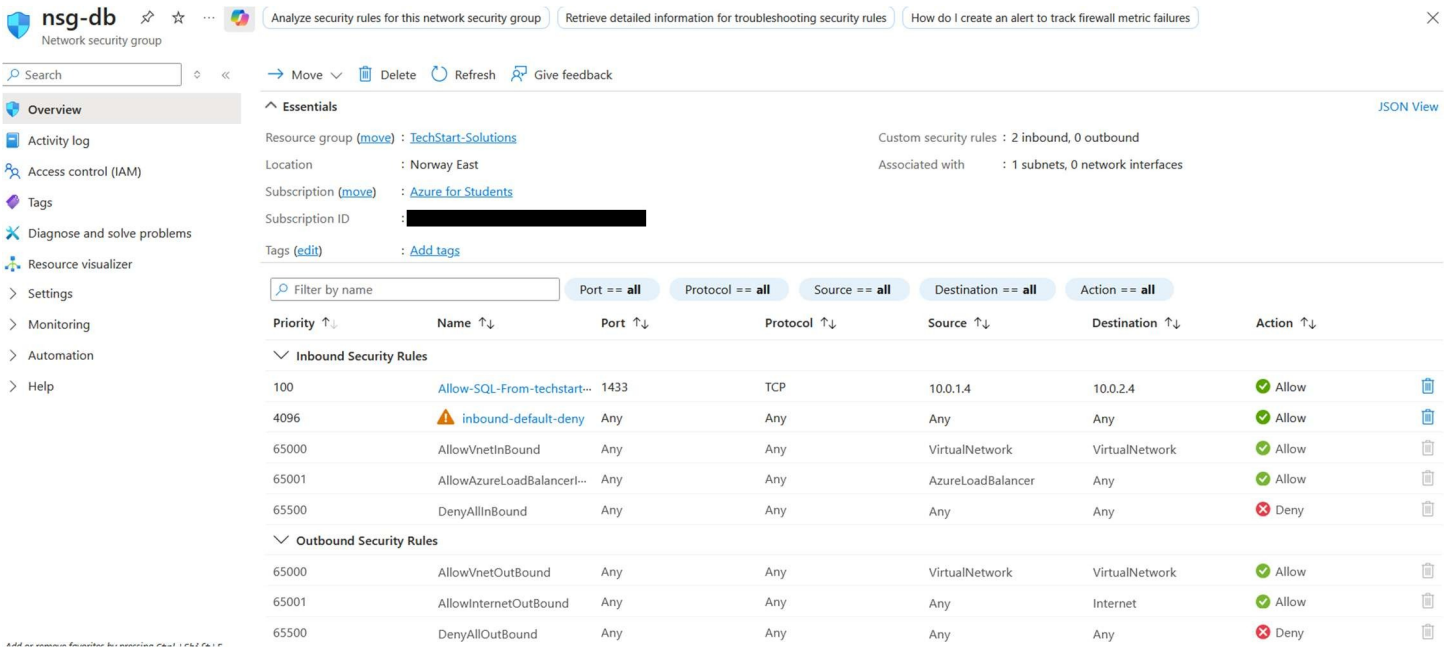


Figure 6 above: nsg-db's security rules that allow only techstart-vm (10.0.1.4) to access the database.

Verification

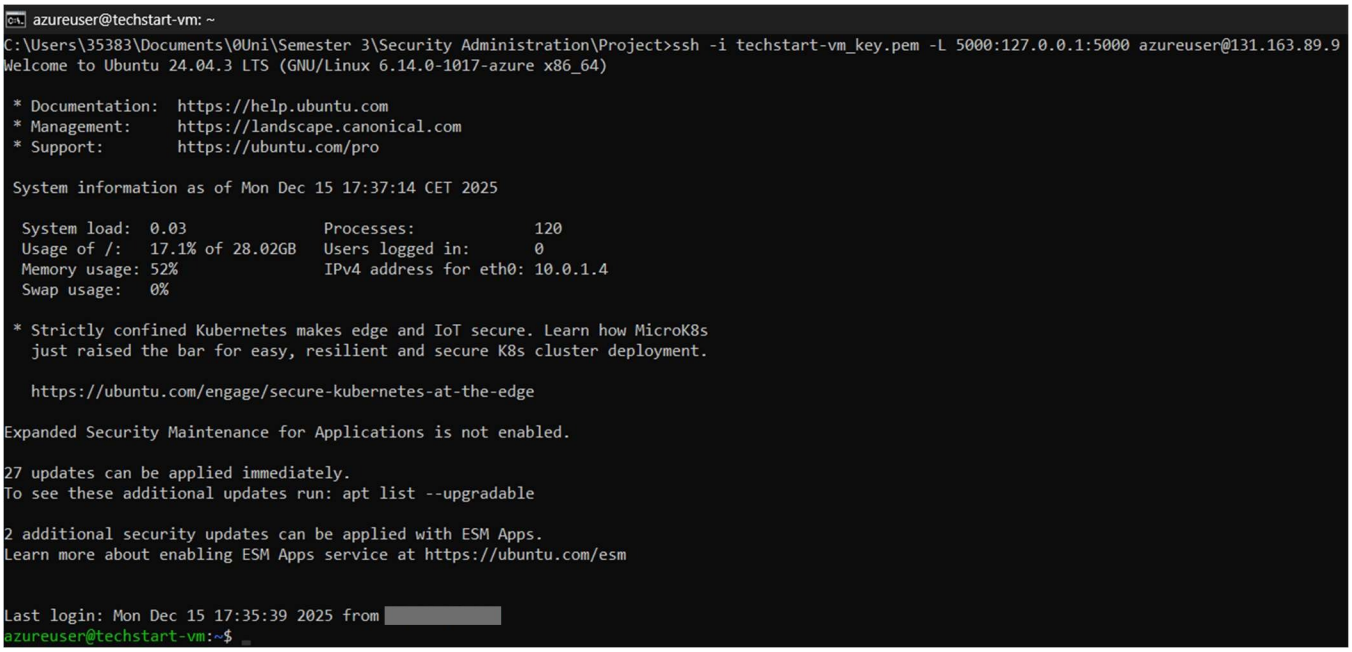


Figure 7 above: SSH connection to techstart-vm's (131.163.89.9) using local port forwarding to tunnel the VM's localhost-only Flask service without exposing it.

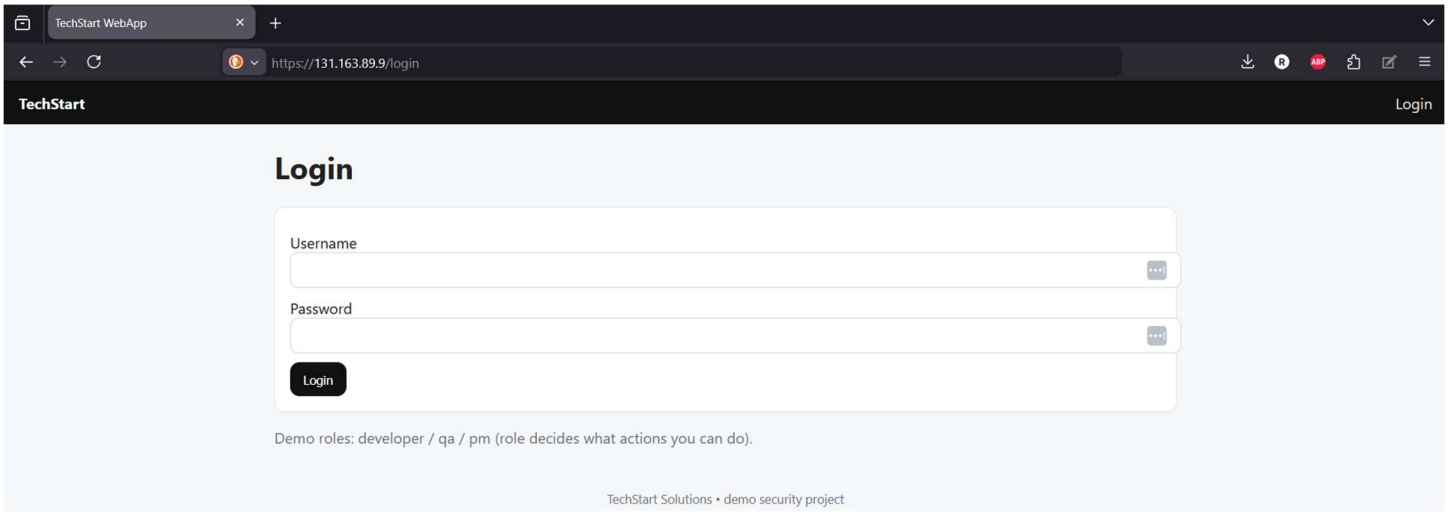


Figure 8 above: HTTPS connection to techstart-vm via web browser. Anyone can access the web application's login page using techstart-vm's public IP address. HTTP connections are allowed but redirected to HTTPS (explained later).

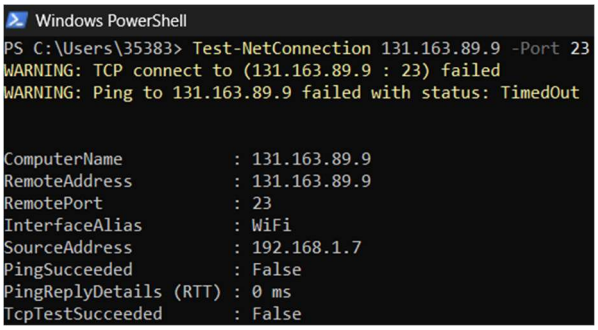


Figure 9 above: no other ports including telnet (TCP 23) can access techstart-vm.

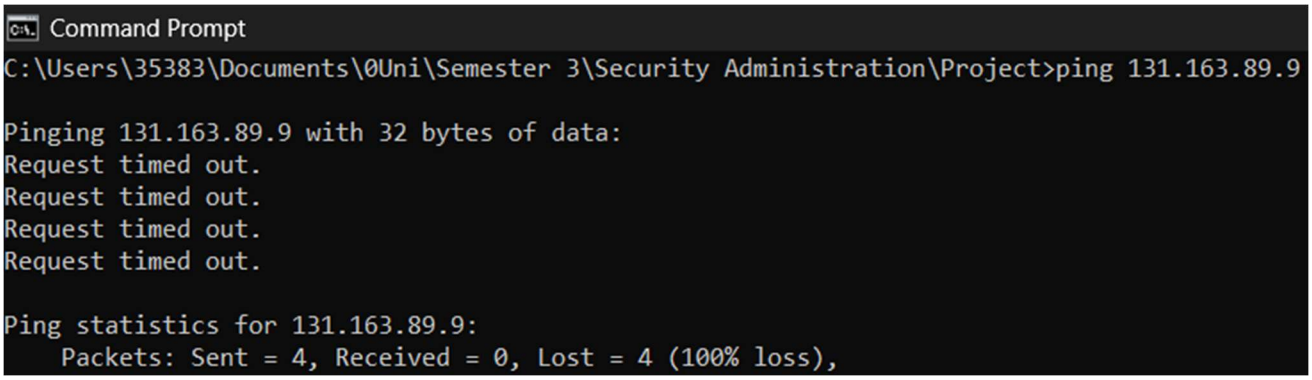


Figure 10 above: incoming ICMP is also blocked which can help mitigate DDoS attacks, specifically ICMP floods.

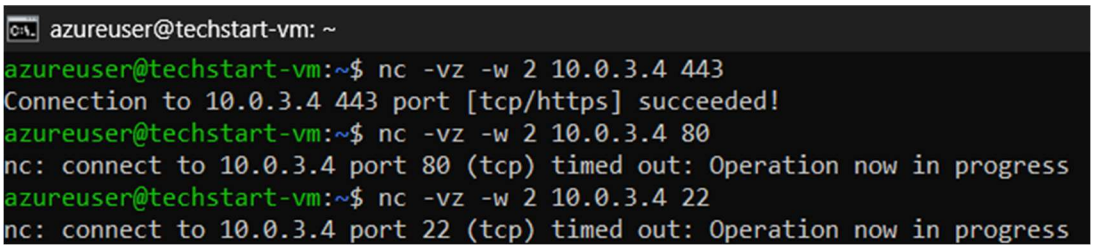


Figure 11 above: HTTPS connections to techstartsolutionstorage's private endpoint's network interface (10.0.3.4) coming from subnet-web (techstart-vm's subnet) are allowed by nsg-storage. Other incoming connections such as HTTP and SSH are denied.

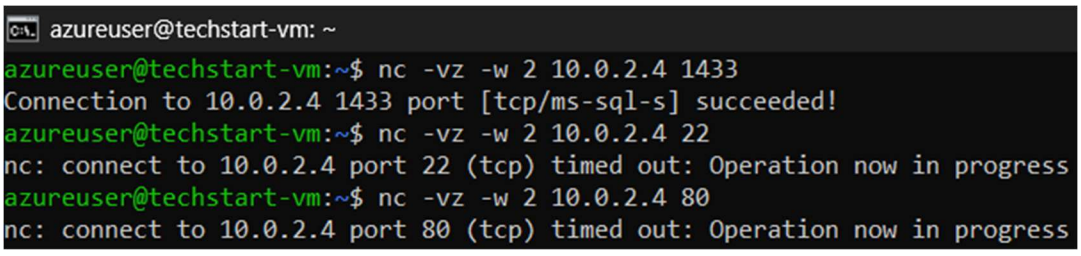


Figure 12 above: Microsoft SQL Server (TCP Port 1433) connections to SQL Server techstartsolutions's private endpoint's network interface (10.0.2.4) coming techstart-vm are allowed by nsg-db. Other incoming connections such as HTTP and SSH are denied.

Strong Authentication and Access Management

Configuration

When techstart-vm was being created on the Azure portal, the authentication type was set to SSH public key instead of password. A Privacy-Enhanced Mail file containing the RSA private key that enables a passwordless authentication for the azureuser account when connecting to techstart-vm via SSH, was generated and downloaded. On techstart-vm, changes were made to the OpenSSH daemon configuration file (/etc/ssh/sshd_config). The lines containing "PubkeyAuthentication yes" and "PasswordAuthentication yes" were uncommented, with the latter's value changed to no, to enable key-based authentication and disable password authentication on techstart-vm. "PermitRootLogin yes" was also uncommented and its value was changed to no to disable root SSH login on techstart-vm.

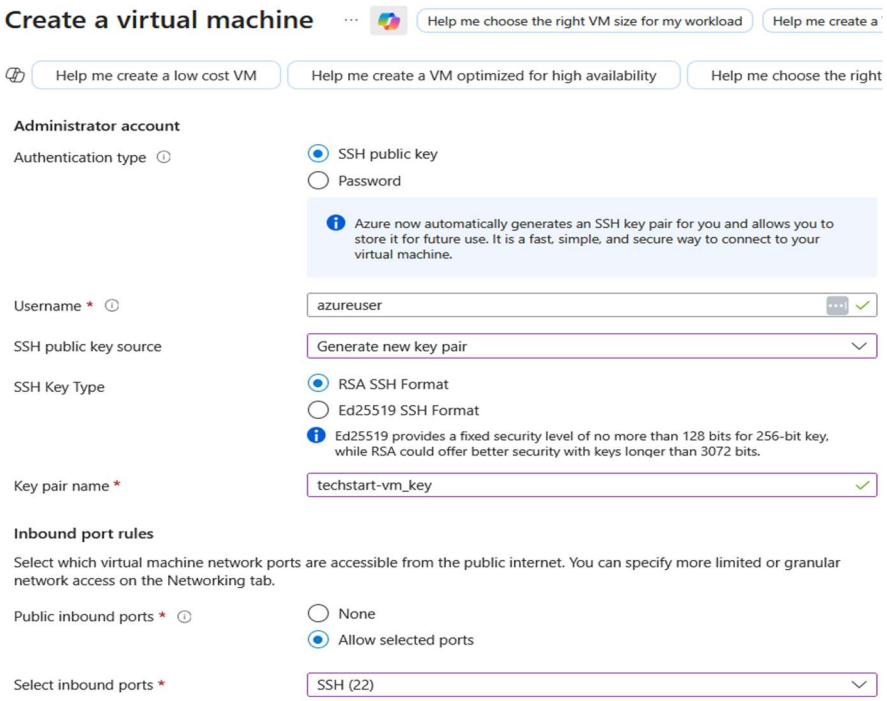
Pluggable Authentication Modules on techstart-vm were configured to enforce account lockout after repeated failed login attempts and password policies. In the /etc/pam.d/common-auth file which manages the validity of authentication attempts, the pam_faillock module was added to first check if the account is locked from five failed authentication attempts within the fail_interval's default previous 15 minutes. If the account is locked, the system does not prompt for credentials for that account. If the authentication attempt fails, the PAM increments the failure counter before locking the account to mitigate against password guessing from unauthorised persons. In the /etc/pam.d/common-password file which applies to user accounts' passwords on the VM, the pam_pwquality module was added to check that passwords being created or changed meets the complexity policies. The pam_pwhistory module was added to ensure that change passwords do not match any of the account's previous five passwords. The pam_unix module was added to store the password using yescrypt hashing.

Strong password policies for users, including the root, on techstart-vm were configured on the /etc/security/pwquality.conf file where minimum requirements are 12 characters in total, containing at least one number, one lowercase character, one uppercase character, and one special character. The passwords cannot be the username and are not able to have more than three consecutive characters that match part of the user nor more three of the same consecutive characters. When changing passwords, four or more characters are required to not have been in the old password. If the inputs of proposed passwords do not meet all of the previously mentioned requirements three times in a row, the system stops prompting for new passwords and returns an error. The /etc/login.defs file contains password aging controls. It was set to require passwords of newly created accounts to be changed no more than every 90 days to mitigate against password guessing, with 14 days notice before the expiration. It also does not allow passwords to be changed again less than seven days after the last change to prevent users who are reluctant to change their current passwords to changing it at least five times to trivial passwords before immediately changing it back to their preferred original one.

On the Azure portal, managed identities were created for all six employees across the three teams working on the web application. Both developers were assigned the most permissive roles out of all the teams, while only having the minimal privileges needed to do their tasks. They are Contributors in TechStart-Solutions RG to create or delete resources in the group, Virtual Machine Contributors in techstart-vm to manage the machine, Storage Blob Data Contributors in techstartsolutionstorage to read, write and delete containers and blobs, and Key Vault Secrets Users in techstart-kv to only read secret contents. Both QAs are Readers in TechStart-Solutions RG to view all resources in the group without being able to make changes, and Storage Blob Data Readers in

techstartsolutionstorage to only read and list containers and blobs. Both Project Managers are also Readers in TechStart-Solutions RG.

Screenshots



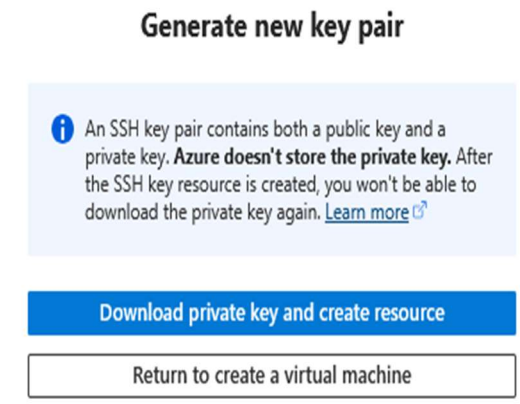


Figure 13 above (left): SSH key-based authentication for techstart-vm's azureuser is enabled (and password authentication disabled) when techstart-vm is being created. The keys are encrypted using RSA and the SSH inbound port is enabled.

Figure 14 above (right): The private SSH key is generated and downloaded in a laptop that manages techstart-vm.

```
azureuser@techstart-vm: /
  GNU nano 7.2                                                    /etc/ssh/sshd_config
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile     .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no
```

Figure 15 above: Password authentication is disabled and key based authentication for users other than the root is enabled in techstart-vm's OpenSSH daemon configuration file to prevent conflicting policies from what was enabled in the Azure portal.

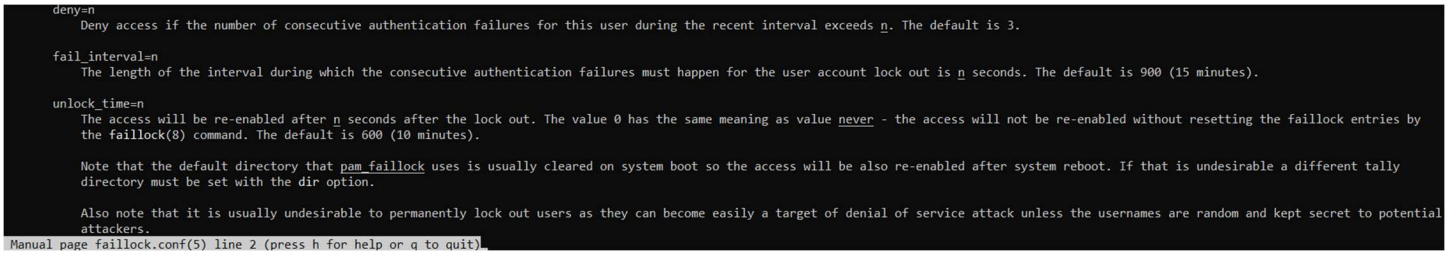


Figure 16 above: as faillock.conf was not configured, the default time that limits the number of failed authentication attempts before the account is locked is 15 minutes, according to the manual.

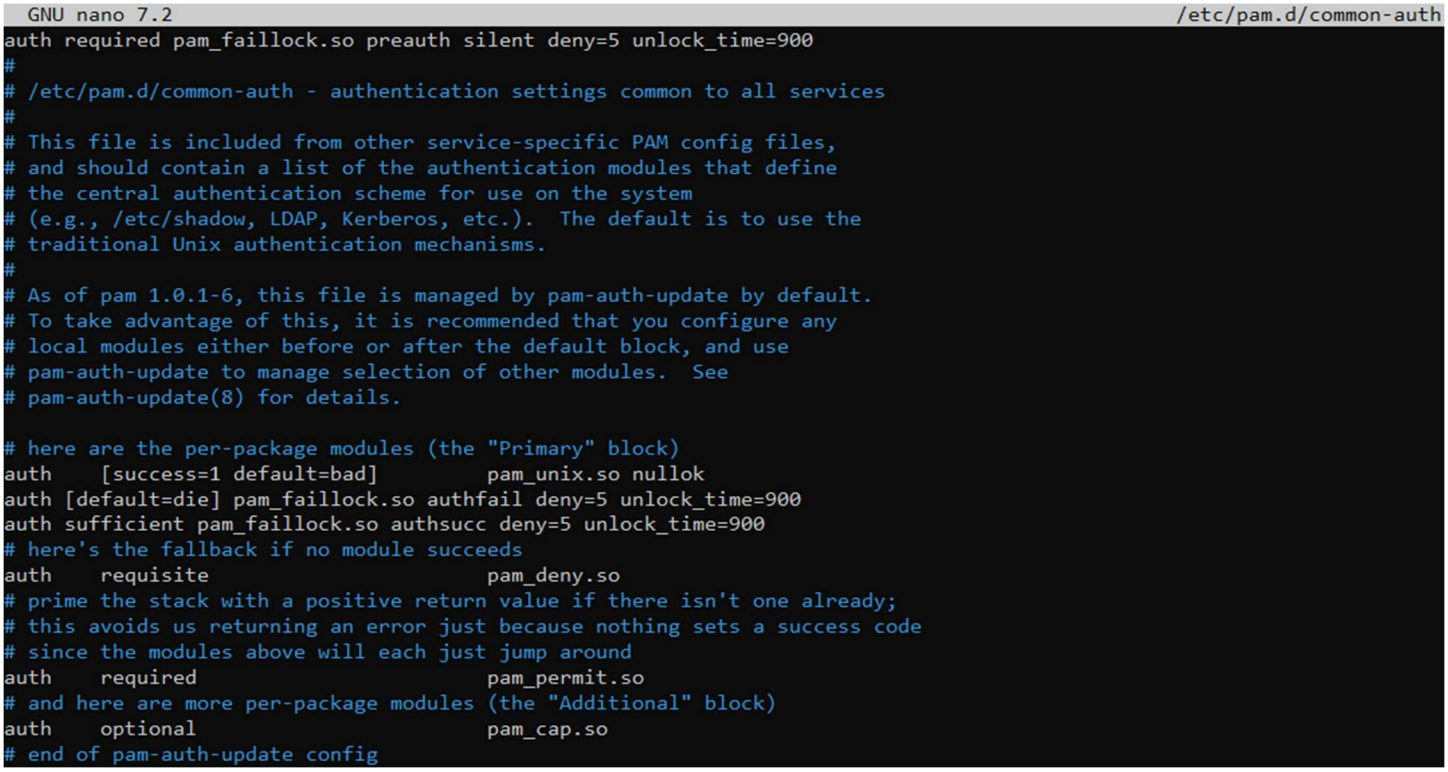


Figure 17 above: the /etc/pam.d/common-auth file which manages all authentications uses the pam_faillock module to check if an authentication attempt for an account already has five failed authentication attempts within the past 15 minutes and to unlock the account after 15 minutes (900 seconds).

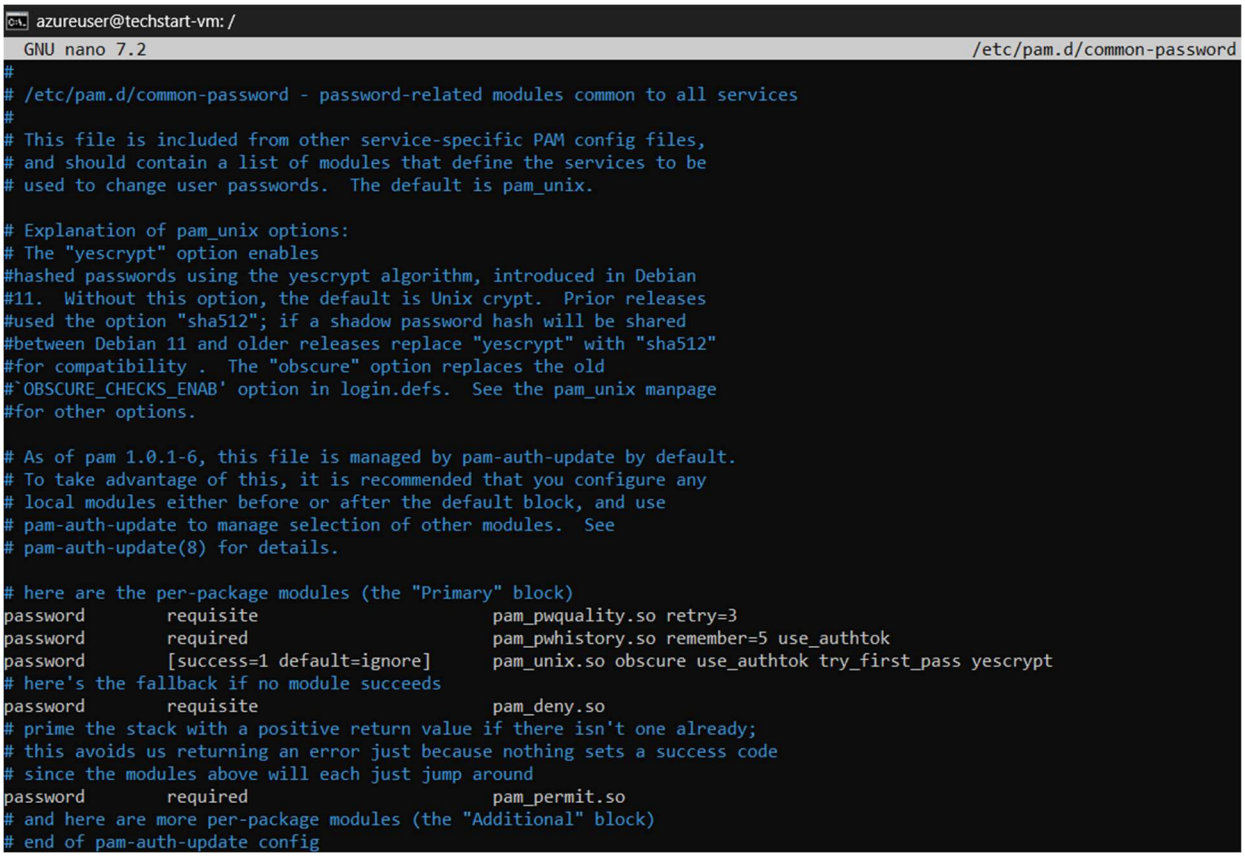


Figure 18 above: the /etc/pam.d/common-password file manages creation and changes of passwords uses the pam_pwquality to enforce complexity rules and pam_pwhistory to refuse the previous 5 passwords being reused. Also uses scrypt hashing.

azureuser@techstart-vm: /

```
  GNU nano 7.2                                                    /etc/security/pwquality.conf
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
difok = 4
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
minlen = 12
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
dcredit = -1
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
ucredit = -1
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
lcredit = -1
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
ocredit = -1
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
# minclass = 0
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
maxrepeat = 3
# Whether to check if it contains the user name in some form.
# The check is enabled if the value is not 0.
usercheck = 1
#
# Length of substrings from the username to check for in the password
# The check is enabled if the value is greater than 0 and usercheck is enabled.
usersubstr = 3
#
# Whether the check is enforced by the PAM module and possibly other
# applications.
# The new password is rejected if it fails the check and the value is not 0.
# enforcing = 1
#
# Path to the cracklib dictionaries. Default is to use the cracklib default.
# dictpath =
#
# Prompt user at most N times before returning with error. The default is 1.
retry = 3
#
# Enforces pwquality checks on the root user password.
# Enabled if the option is present.
enforce_for_root
```

Figure 19 above: the /etc/security/pwquality.conf file sets the complexity rules when a password is being created or changed such as requiring a minimum amount of certain characters.

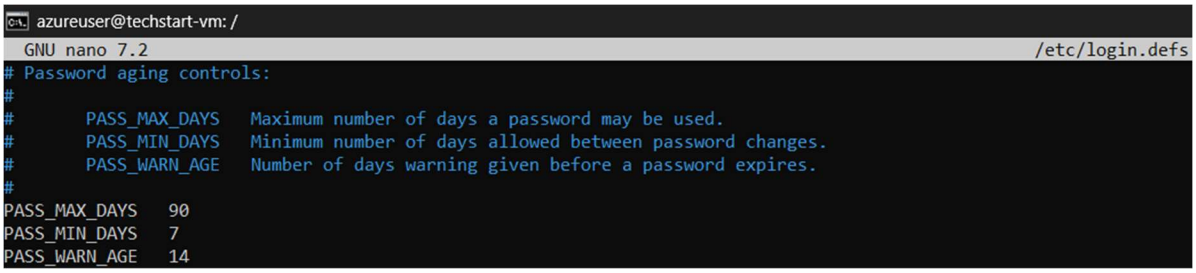


Figure 20 above: the /etc/login.defs file manages when passwords can be changed and when they must be changed.

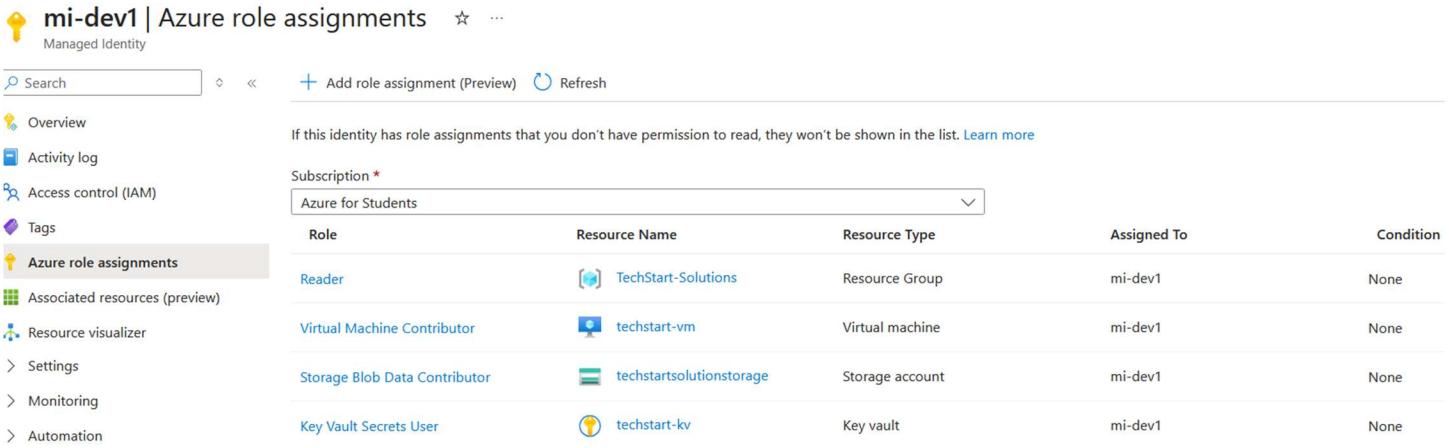


Figure 21 above: dev1's (and dev2) RBAC have the most powerful privileges in the most amount of resources compared to other employees in order to do their tasks while following the Principle of Least Privilege.

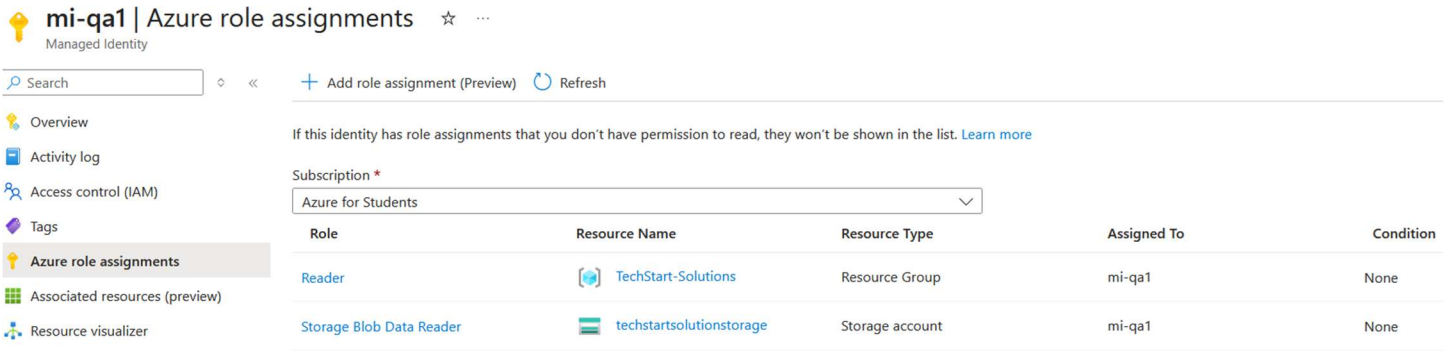


Figure 22 above: qa1's (and qa2) RBAC has less privileges as their job is to test the web application rather than manage it.

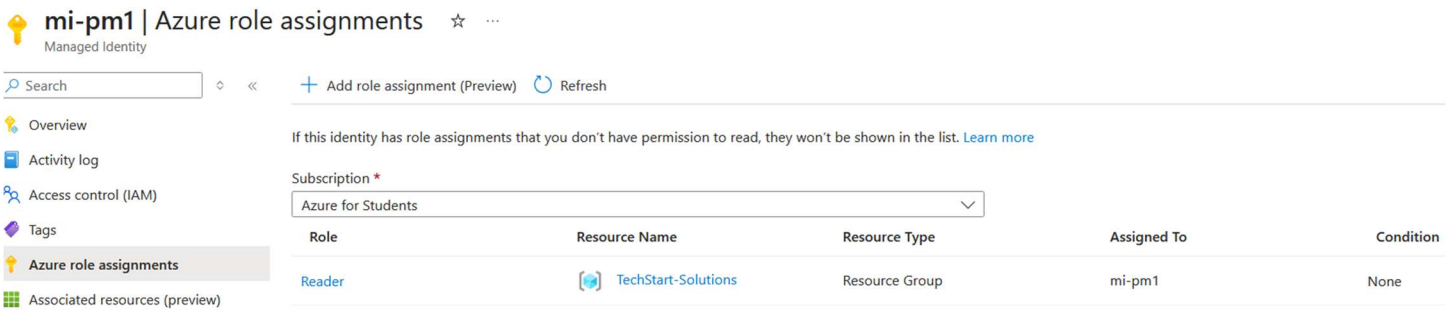


Figure 23 above: pm1's (and pm2) RBAC has the least amount of privileges, only being allowed to view the resources in the project.

Verification

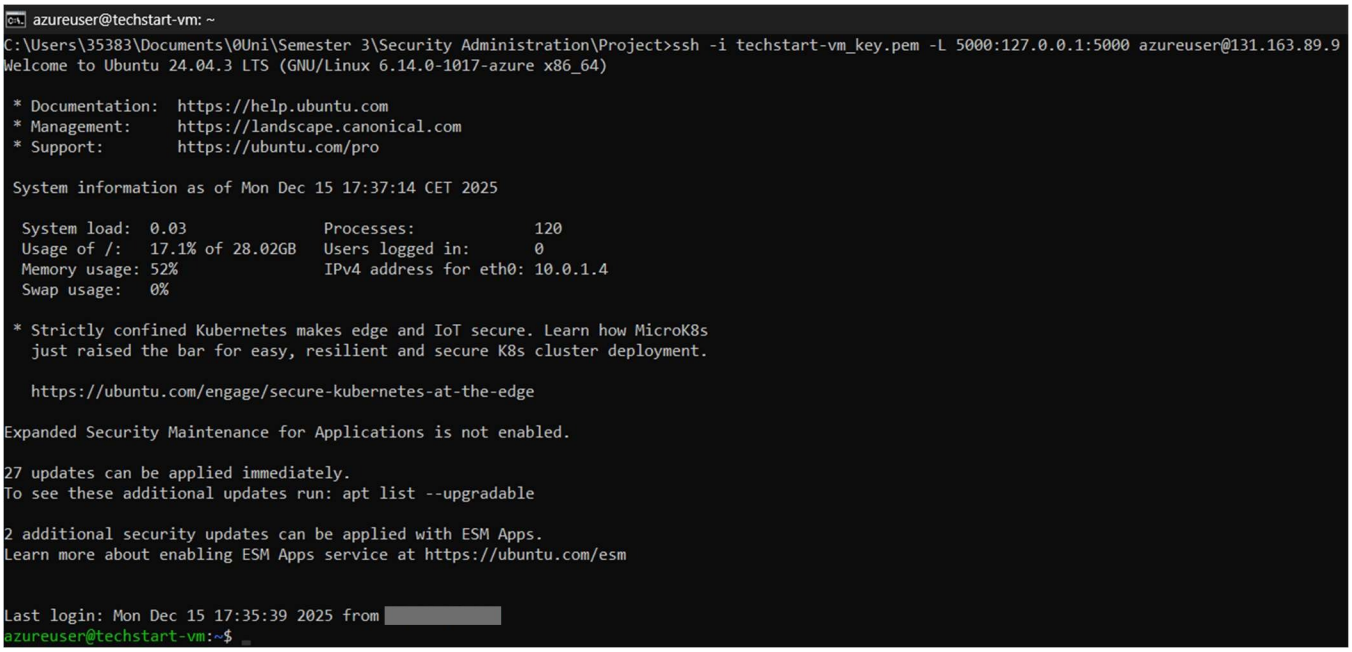


Figure 24 above: successful authentication to techstart-vm's azureuser requires the Privacy-Enhanced Mail file containing the private key.

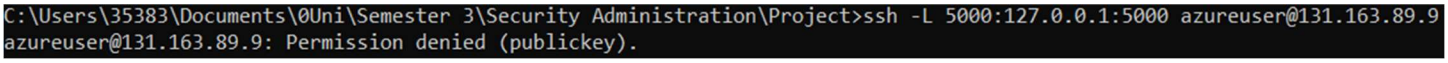


Figure 25 above: access to techstart-vm cannot be authenticated using passwords or any other way besides using a key.

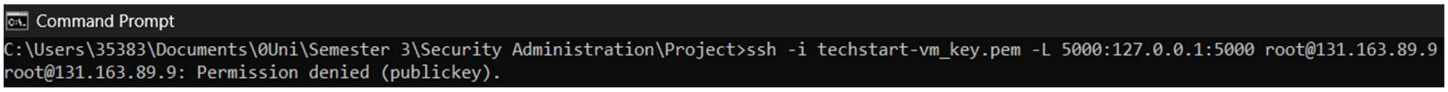


Figure 26 above: Root login using SSH is disabled even if using the PEM key file (which can only be used to authenticate azureuser).

```
azureuser@techstart-vm:~$ sudo useradd -m locktest
azureuser@techstart-vm:~$ sudo usermod -s /bin/bash locktest
azureuser@techstart-vm:~$ sudo passwd locktest
New password:
Retype new password:
passwd: password updated successfully
azureuser@techstart-vm:~$ su locktest
Password:
su: Authentication failure
azureuser@techstart-vm:~$ su locktest
Password:
su: Authentication failure
azureuser@techstart-vm:~$ su locktest
Password:
su: Authentication failure
azureuser@techstart-vm:~$ su locktest
Password:
su: Authentication failure
azureuser@techstart-vm:~$ su locktest
Password:
su: Authentication failure
azureuser@techstart-vm:~$ su locktest
su: Authentication failure
azureuser@techstart-vm:~$ su locktest
Password:
locktest@techstart-vm:/home/azureuser$ exit
exit
azureuser@techstart-vm:~$ sudo tail -n 30 /var/log/auth.log | egrep -i 'locktest|FAILED SU|authentication failure'
2025-12-16T21:13:05.591367+01:00 techstart-vm sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/bin/passwd locktest
2025-12-16T21:13:09.476387+01:00 techstart-vm passwd[60294]: pam_unix(passwd:chauthtok): password changed for locktest
2025-12-16T21:13:20.186255+01:00 techstart-vm su: pam_unix(su:auth): authentication failure; logname=azureuser uid=1000 euid=0 tty=/dev/pts/0 ruser=azureuser rhost=  user=locktest
2025-12-16T21:13:22.333202+01:00 techstart-vm su[60296]: FAILED SU (to locktest) azureuser on pts/0
2025-12-16T21:13:27.067249+01:00 techstart-vm su: pam_unix(su:auth): authentication failure; logname=azureuser uid=1000 euid=0 tty=/dev/pts/0 ruser=azureuser rhost=  user=locktest
2025-12-16T21:13:29.039748+01:00 techstart-vm su[60297]: FAILED SU (to locktest) azureuser on pts/0
2025-12-16T21:13:35.469023+01:00 techstart-vm su: pam_unix(su:auth): authentication failure; logname=azureuser uid=1000 euid=0 tty=/dev/pts/0 ruser=azureuser rhost=  user=locktest
2025-12-16T21:13:37.009228+01:00 techstart-vm su[60298]: FAILED SU (to locktest) azureuser on pts/0
2025-12-16T21:13:41.049327+01:00 techstart-vm su: pam_unix(su:auth): authentication failure; logname=azureuser uid=1000 euid=0 tty=/dev/pts/0 ruser=azureuser rhost=  user=locktest
2025-12-16T21:13:42.608932+01:00 techstart-vm su[60299]: FAILED SU (to locktest) azureuser on pts/0
2025-12-16T21:13:47.375626+01:00 techstart-vm su: pam_unix(su:auth): authentication failure; logname=azureuser uid=1000 euid=0 tty=/dev/pts/0 ruser=azureuser rhost=  user=locktest
2025-12-16T21:13:47.376077+01:00 techstart-vm su: pam_faillock(su:auth): Consecutive login failures for user locktest account temporarily locked
2025-12-16T21:13:48.896659+01:00 techstart-vm su[60300]: FAILED SU (to locktest) azureuser on pts/0
2025-12-16T21:13:54.768391+01:00 techstart-vm su[60301]: FAILED SU (to locktest) azureuser on pts/0
2025-12-16T21:29:30.971947+01:00 techstart-vm su[60344]: (to locktest) azureuser on pts/0
2025-12-16T21:29:30.979831+01:00 techstart-vm su[60344]: pam_unix(su:session): session opened for user locktest(uid=1013) by azureuser(uid=1000)
2025-12-16T21:29:46.862941+01:00 techstart-vm su[60344]: pam_unix(su:session): session closed for user locktest
```

Figure 27 above: a new user "locktest" is created, its shell is changed to bash for consistency when switching users, and a password was set. Incorrect passwords were given 5 times in a row, locking the user for 15 minutes which made the system stop prompting for the password and returned an Authentication failure in the sixth attempt. Authentication was possible after 15 minutes.

```
azureuser@techstart-vm:~$ sudo useradd -m passtest
azureuser@techstart-vm:~$ sudo usermod -s /bin/bash passtest
azureuser@techstart-vm:~$ sudo passwd passtest
New password:
Retype new password:
passwd: password updated successfully
azureuser@techstart-vm:~$ su passtest
Password:
passtest@techstart-vm:/home/azureuser$ passwd passtest
Changing password for passtest.
Current password:
New password:
BAD PASSWORD: The password contains less than 1 digits
New password:
BAD PASSWORD: The password contains less than 1 uppercase letters
New password:
BAD PASSWORD: The password contains less than 1 lowercase letters
passwd: Have exhausted maximum number of retries for service
passwd: password unchanged
passtest@techstart-vm:/home/azureuser$ passwd passtest
Changing password for passtest.
Current password:
New password:
BAD PASSWORD: The password contains less than 1 non-alphanumeric characters
New password:
BAD PASSWORD: The password is shorter than 12 characters
New password:
BAD PASSWORD: The password contains more than 3 same characters consecutively
passwd: Have exhausted maximum number of retries for service
passwd: password unchanged
passtest@techstart-vm:/home/azureuser$ passwd passtest
Changing password for passtest.
Current password:
New password:
BAD PASSWORD: The password contains the user name in some form
New password:
BAD PASSWORD: The password is too similar to the old one
New password:
Retype new password:
passwd: password updated successfully
```

Figure 28 above: a new user "passtest" is created, its shell is changed to bash for consistency when switching users, and a password was set. After switching to passtest (doing it on azureuser would override the password policies), multiple attempts to change the current password to different new passwords that each failed to meet a different policy were denied, until a new password that met all of the policies was given.

```
azureuser@techstart-vm:~$ sudo useradd -m agetest
azureuser@techstart-vm:~$ sudo usermod -s /bin/bash agetest
azureuser@techstart-vm:~$ sudo passwd agetest
New password:
Retype new password:
passwd: password updated successfully
azureuser@techstart-vm:~$ su agetest
Password:
agetest@techstart-vm:/home/azureuser$ passwd agetest
Changing password for agetest.
Current password:
You must wait longer to change your password.
passwd: Authentication token manipulation error
passwd: password unchanged
```

Figure 29 above: after setting the password aging controls, a new user "agetest" is created, its shell is changed to bash for consistency when switching users, and a password was set. After immediately switching to agetest (doing it on azureuser would override the password policies), an attempt to change the password was unsuccessful as users must wait a week.

```
azureuser@techstart-vm:~$ az login --identity --client-id 56654d7e-6ec9-4fbd-80e5-4d2d730c3e10 -o none
azureuser@techstart-vm:~$ az identity show -g TechStart-Solutions -n mi-dev1 --query "{identityName:name, clientId:clientId, principalId:principalId}" -o table
IdentityName    ClientId                              PrincipalId
--------------  ------------------------------------  ------------------------------------
mi-dev1         56654d7e-6ec9-4fbd-80e5-4d2d730c3e10  94894003-b9c2-4b5b-bd1c-2c53df29a176
azureuser@techstart-vm:~$ az resource list -g "TechStart-Solutions" --query "[].{name:name}" -o table
Name
------------------------------------------------------------
techstartsolutions
techstartsolutions/master
techstartsolutions/TechStartDB
techstart-vm_key
techstart-vm-nsg
techstart-vm-ip
techstart-vnet
techstart-vm239_z1
techstart-vm
techstart-vm_OsDisk_1_e61b475a101045d6a6560a7bc5b906ef
sql-privateendpoint
sql-privateendpoint-nic
privatelink.database.windows.net
privatelink.database.windows.net/4y3xzr4wlzpdc
techstartsolutionstorage
blob-pe
blob-pe-nic
privatelink.blob.core.windows.net
privatelink.blob.core.windows.net/4y3xzr4wlzpdc
nsg-storage
mi-projectmanagers
mi-developers
mi-qa
techstart-kv
key-privateendpoint
key-privateendpoint.nic.61c12432-952e-471b-837b-1159cff0db55
privatelink.vaultcore.azure.net
privatelink.vaultcore.azure.net/4y3xzr4wlzpdc
mi-dev1
mi-dev2
mi-qa1
mi-qa2
mi-pm1
mi-pm2
nsg-db
azureuser@techstart-vm:~$ az vm update -g TechStart-Solutions -n techstart-vm --set tags.RBAC_Test=dev1 -o none
azureuser@techstart-vm:~$ az vm show -g TechStart-Solutions -n techstart-vm --query "{name:name, RBAC_Test:tags.RBAC_Test}" -o table
Name          RBAC_Test
------------  -----------
techstart-vm  dev1
azureuser@techstart-vm:~$ echo "test" > /tmp/dev1.txt
azureuser@techstart-vm:~$ az storage blob upload --account-name techstartsolutionstorage --container-name app-files --name dev1-verify.txt --file /tmp/dev1.txt --auth-mode login --overwrite -o none
Finished[#############################################################]  100.0000%
azureuser@techstart-vm:~$ az storage blob list --account-name techstartsolutionstorage --container-name app-files --auth-mode login --query "[].{Name:name, Container:container}" -o table
Name             Container
---------------  -----------
dev1-verify.txt  app-files
```

Figure 30 above: logging into dev1's managed identity using its client-id, dev1 (and dev2) can read the resources in TechStart-Solutions Resource Group, can add and read a VM tag, and add and read a blob file on techstartsolutionstorage.

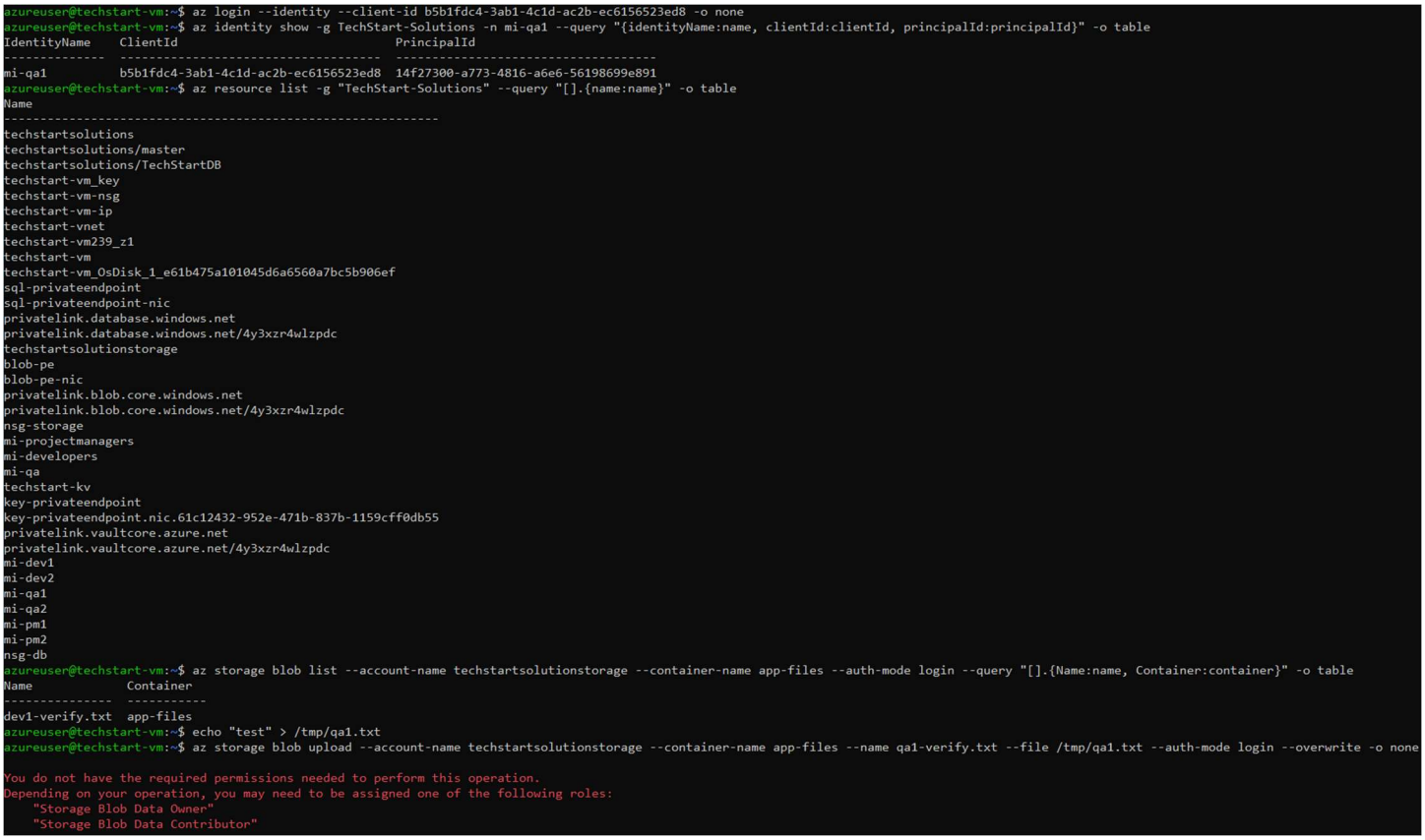


Figure 31 above: logging into qa1's managed identity using its client-id, qa1 (and qa2) can read the resources in TechStart-Solutions Resource Group and read blob files on techstartsolutionstorage but cannot add a blob file.

```
azureuser@techstart-vm:~$ az login --identity --client-id 493c3e8d-24f9-448a-a3d5-e8381b151d3d -o none
azureuser@techstart-vm:~$ az identity show -g TechStart-Solutions -n mi-pm1 --query "{identityName:name, clientId:clientId, principalId:principalId}" -o table
IdentityName    ClientId                              PrincipalId
--------------  ------------------------------------  ------------------------------------
mi-pm1          493c3e8d-24f9-448a-a3d5-e8381b151d3d  823828c6-c475-4abc-916f-03229b8d6ae5
azureuser@techstart-vm:~$ az resource list -g "TechStart-Solutions" --query "[].{name:name}" -o table
Name
------------------------------------------------------------
techstartsolutions
techstartsolutions/master
techstartsolutions/TechStartDB
techstart-vm_key
techstart-vm-nsg
techstart-vm-ip
techstart-vnet
techstart-vm239_z1
techstart-vm
techstart-vm_OsDisk_1_e61b475a101045d6a6560a7bc5b906ef
sql-privateendpoint
sql-privateendpoint-nic
privatelink.database.windows.net
privatelink.database.windows.net/4y3xzr4wlzpdc
techstartsolutionstorage
blob-pe
blob-pe-nic
privatelink.blob.core.windows.net
privatelink.blob.core.windows.net/4y3xzr4wlzpdc
nsg-storage
mi-projectmanagers
mi-developers
mi-qa
techstart-kv
key-privateendpoint
key-privateendpoint.nic.61c12432-952e-471b-837b-1159cff0db55
privatelink.vaultcore.azure.net
privatelink.vaultcore.azure.net/4y3xzr4wlzpdc
mi-dev1
mi-dev2
mi-qa1
mi-qa2
mi-pm1
mi-pm2
nsg-db
azureuser@techstart-vm:~$ az storage blob list --account-name techstartsolutionstorage --container-name app-files --auth-mode login --query "[].{Name:name, Container:container}" -o table

You do not have the required permissions needed to perform this operation.
Depending on your operation, you may need to be assigned one of the following roles:
    "Storage Blob Data Owner"
    "Storage Blob Data Contributor"
    "Storage Blob Data Reader"
    "Storage Queue Data Contributor"
    "Storage Queue Data Reader"
    "Storage Table Data Contributor"
    "Storage Table Data Reader"
```

Figure 32 above: logging into pm1's managed identity using its client-id, pm1 (and pm2) can only read the resources in TechStart-Solutions Resource Group. It cannot do anything else such as reading blob files on techstartsolutionstorage.

Encryption in Transit and at Rest

Configuration

On the Azure portal, Transparent Data Encryption, specifically Service-managed key which is automatically generated by Azure to encrypt the database, is enabled on the techstartsolutions SQL server. Customer-managed key was considered as it gave more control over key management tasks, but ultimately it was not implemented as it adds an accessibility risk to the database if the key vault key is missing. TDE is also enabled on the TechStartDB SQL Database to ensure encryption at rest for all database files such as data, log, and backups.

Encryption at rest in Azure Blob Storage is enabled by using techstartsolutionstorage's service-side encryption, which encrypts all data written to techstartsolutionstorage automatically by using Microsoft-managed keys. This control protects data on disk without requiring application changes and ensures that the blob contents stored in techstartsolutionstorage remains encrypted, even if the storage media were accessed outside of Azure. Customer-managed key was also considered for more control over key management tasks, but once again it was not implemented due to the accessibility risk.

SQL connections were configured to use TLS encryption to protect data in transit between techstart-vm and TechStartDB with a Minimum TLS version set on the techstartsolutions SQL server that prevents older versions than 1.2 from connecting.

HTTPS was enabled on the Flask CRUD application by configuring Apache with mod_ssl and a self-signed TLS certificate on port 443, to protect user credentials and session cookies in transit by encrypting them. A free self-signed certificate was used because the site is accessed directly by techstart-vm's public IP address with no DNS to validate with a trusted Certificate Authority. However, the self-signed certificate does trigger a browser warning that requires users to confirm that they wish to continue before being able to use the web application. Apache was then configured so that the HTTP virtual host on port 80 only performs an HTTP to HTTPS redirect, to ensure that users who type the IP address are pushed onto an encrypted connection by default.

Screenshots

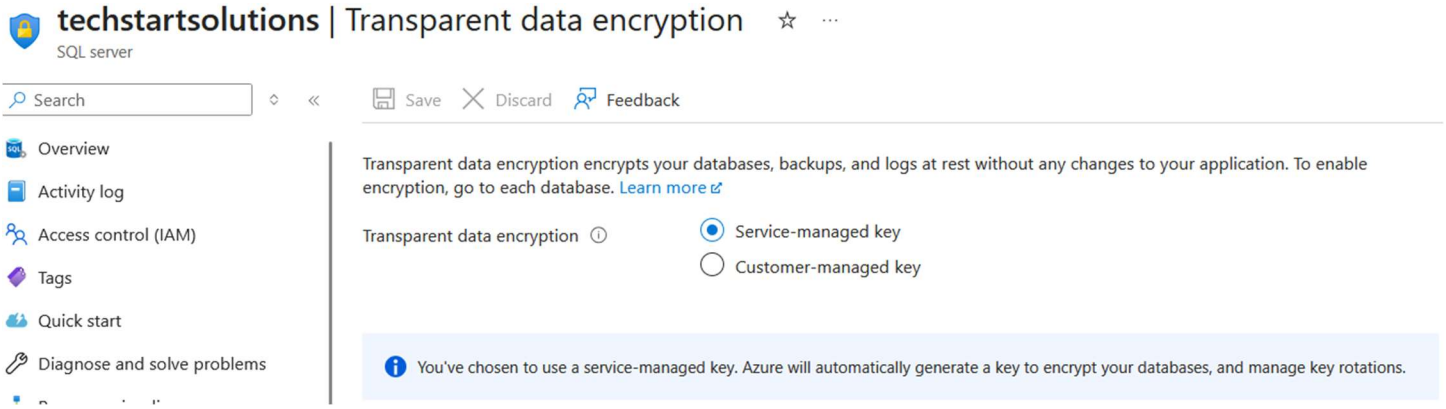


Figure 33 above: Transparent data encryption is enabled on techstartsolutions SQL server and uses a service-managed key.

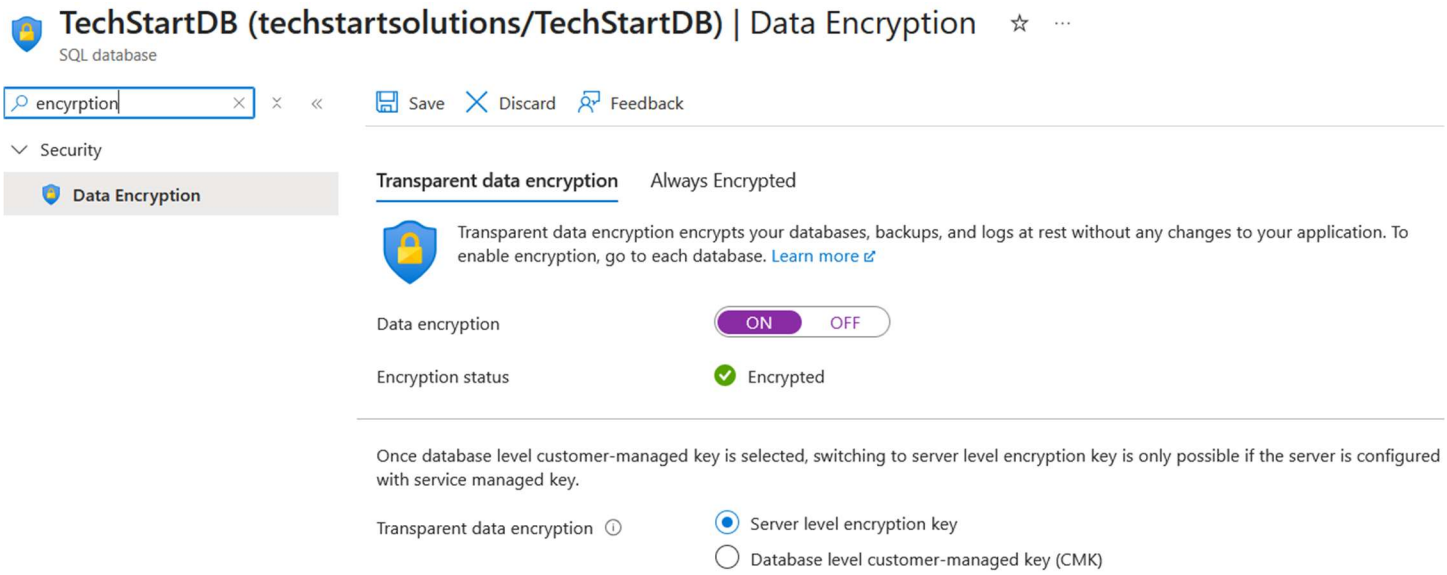


Figure 34 above: Transparent data encryption is enabled on TechStartDB and uses a server level encryption key.

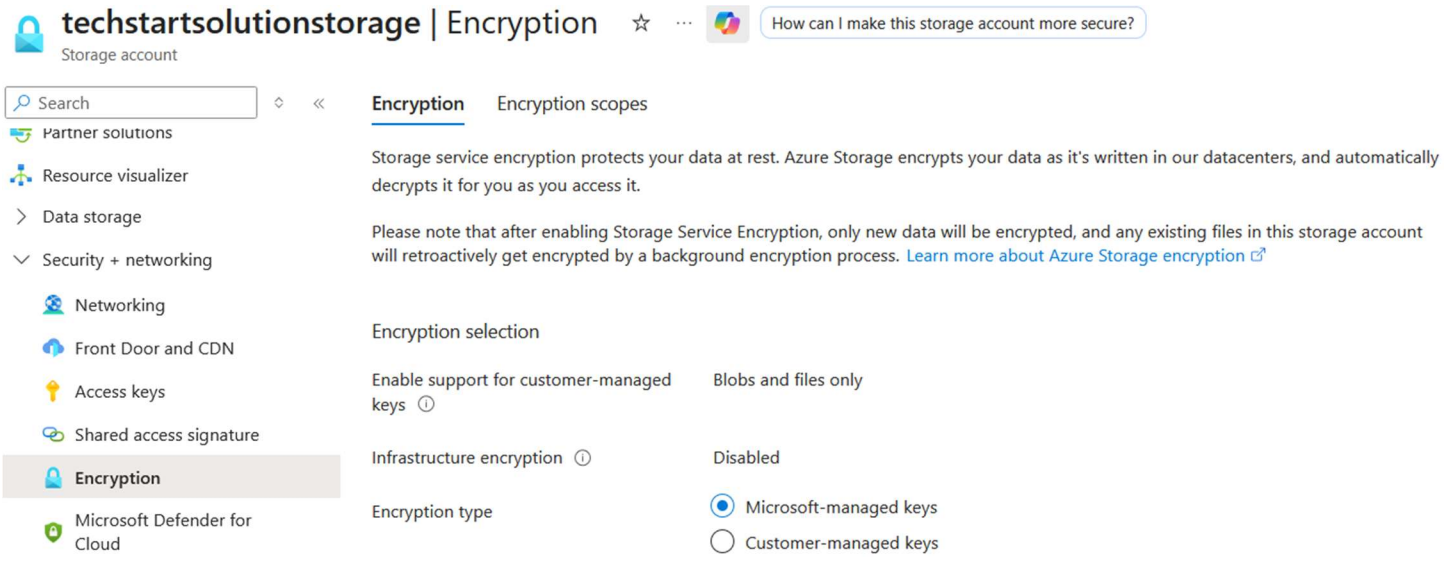


Figure 35 above: Encryption at rest is enabled on techstartsolutionstorage and uses Microsoft-managed keys.

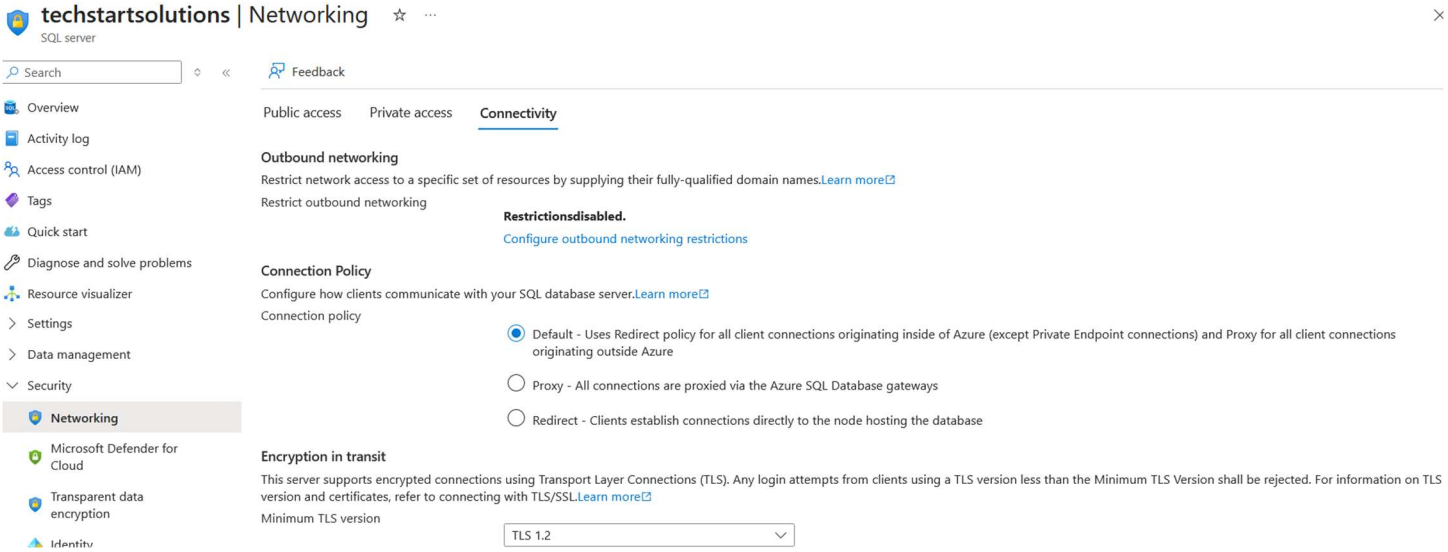


Figure 36 above: TLS encryption (version 1.2 or later) is required for connections with techstartsolutions SQL server.

```
  GNU nano 7.2                                          /etc/apache2/sites-available/flaskapp-ssl.conf
<VirtualHost *:443>
    ServerName 131.163.89.9

    SSLEngine on
    SSLCertificateFile /etc/apache2/ssl/techstart.crt
    SSLCertificateKeyFile /etc/apache2/ssl/techstart.key

    WSGIDaemonProcess flaskapp_ssl python-home=/var/www/flaskapp/venv python-path=/var/www/flaskapp
    WSGIProcessGroup flaskapp_ssl
    WSGIScriptAlias / /var/www/flaskapp/flaskapp.wsgi

    <Directory /var/www/flaskapp>
        Require all granted
    </Directory>

    Alias /static /var/www/flaskapp/static
    <Directory /var/www/flaskapp/static>
        Require all granted
    </Directory>

    SetEnv KEY_VAULT_NAME techstart-kv
    SetEnv BLOB_CONTAINER_NAME app-files
    SetEnv SQL_SECRET_NAME sql-connection-string
    SetEnv STORAGE_SECRET_NAME storage-connection-string
    SetEnv FLASK_SECRET_KEY

    ErrorLog ${APACHE_LOG_DIR}/flaskapp_ssl_error.log
    CustomLog ${APACHE_LOG_DIR}/flaskapp_ssl_access.log combined
</VirtualHost>
```

Figure 37 above: Apache HTTPS VirtualHost on port 443 is configured on techstart-vm's flaskapp-ssl.conf file, enabling TLS with a self-signed certificate and serving the Flask CRUD application using mod_wsgi.

```
azureuser@techstart-vm:~$ sudo nano /etc/apache2/sites-available/flaskapp-ssl.conf
azureuser@techstart-vm:~$ sudo a2ensite flaskapp-ssl.conf
Enabling site flaskapp-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
azureuser@techstart-vm:~$ sudo systemctl reload apache2
```

Figure 38 above: the HTTPS site configuration on flaskapp-ssl.conf is enabled with a2ensite and applied by reloading Apache.

```
  GNU nano 7.2                                                    /etc/apache2/sites-available/flaskapp.conf
<VirtualHost *:80>
    ServerName 131.163.89.9
    RewriteEngine On
    RewriteRule ^ https://%{HTTP_HOST}%{REQUEST_URI} [L,R=301]
    WSGIDaemonProcess flaskapp python-home=/var/www/flaskapp/venv python-path=/var/www/flaskapp
    WSGIProcessGroup flaskapp
    WSGIScriptAlias / /var/www/flaskapp/flaskapp.wsgi
    <Directory /var/www/flaskapp>
        Require all granted
    </Directory>
    Alias /static /var/www/flaskapp/static
    <Directory /var/www/flaskapp/static>
        Require all granted
    </Directory>
    SetEnv KEY_VAULT_NAME techstart-kv
    SetEnv BLOB_CONTAINER_NAME app-files
    SetEnv SQL_SECRET_NAME sql-connection-string
    SetEnv STORAGE_SECRET_NAME storage-connection-string
    SetEnv FLASK_SECRET_KEY [redacted]
    ErrorLog ${APACHE_LOG_DIR}/flaskapp_error.log
    CustomLog ${APACHE_LOG_DIR}/flaskapp_access.log combined
</VirtualHost>
```

Figure 39 above: Apache HTTP VirtualHost on port 80 is configured on flaskapp.conf to permanently redirect (code 301) all HTTP requests to HTTPS.

Verification

```
azureuser@techstart-vm:~$ sqlcmd -S techstartsolutions.database.windows.net -d TechStartDB -U sqladmin -N -C -y 20 -Y 20
Password:
1> SELECT
2>      DB_NAME(database_id) AS database_name,
3>      encryption_state,
4>      CASE encryption_state
5>              WHEN 0 THEN 'No database encryption key present'
6>              WHEN 1 THEN 'Unencrypted'
7>              WHEN 2 THEN 'Encryption in progress'
8>              WHEN 3 THEN 'Encrypted'
9>              WHEN 4 THEN 'Key change in progress'
10>             WHEN 5 THEN 'Decryption in progress'
11>             WHEN 6 THEN 'Protection change in progress'
12>     END AS encryption_state_desc,
13>     key_algorithm,
14>     key_length
15> FROM sys.dm_database_encryption_keys;
16> GO
database_name        encryption_state encryption_state_des key_algorithm        key_length
-------------------- ---------------- -------------------- -------------------- -----------
tempdb                             3 Encrypted            AES                          256
TechStartDB                        3 Encrypted            AES                          256

(2 rows affected)
```

Figure 40 above: SQL is used to query the database's key details, the encryption state, and its corresponding description to verify TDE is enabled.

```
azureuser@techstart-vm:/$ echo "encryption-test" > /tmp/enc-test.txt
azureuser@techstart-vm:/$ az storage blob upload --account-name techstartsolutionstorage --container-name app-files --name enc-proof.txt --file /tmp/enc-test.txt --auth-mode login --overwrite -o jsonc
Finished[#############################################################]  100.0000%
{
  "client_request_id": "f5af1ef3-de0f-11f0-9d30-7ced8d254f89",
  "content_md5": "w4LwRDfXWqVieXyFpAqUrw==",
  "date": "2025-12-17T01:54:17+00:00",
  "encryption_key_sha256": null,
  "encryption_scope": null,
  "etag": "\"0x8DE4033DA5AA866\"",
  "lastModified": "2025-12-17T01:54:18+00:00",
  "request_id": "61cac1bb-901e-0008-3e1c-727bd1000000",
  "request_server_encrypted": true,
  "version": "2022-11-02",
  "version_id": null
}
```

Figure 41 above: an upload of a blob to app-files container shows encryption at rest on techstartsolutionstorage is enabled.

To verify the SQL traffic is encrypted in transit over TLS 1.2:

Output of: openssl s_client -connect techstartsolutions.database.windows.net:1433 -tls1_2 < /dev/null:

```
CONNECTED(00000003)
depth=2 C = US, O = DigiCert Inc, OU = www.digicert.com, CN = DigiCert Global Root G2
verify return:1
depth=1 C = US, O = Microsoft Corporation, CN = Microsoft Azure RSA TLS Issuing CA 04
verify return:1
depth=0 C = US, ST = WA, L = Redmond, O = Microsoft Corporation, CN = cr15.norwayeast1-
a.control.database.windows.net
verify return:1
---
Certificate chain
 0 s:C = US, ST = WA, L = Redmond, O = Microsoft Corporation, CN = cr15.norwayeast1-
a.control.database.windows.net
   i:C = US, O = Microsoft Corporation, CN = Microsoft Azure RSA TLS Issuing CA 04
   a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA384
   v:NotBefore: Oct 19 11:26:57 2025 GMT; NotAfter: Apr 17 11:26:57 2026 GMT
 1 s:C = US, O = Microsoft Corporation, CN = Microsoft Azure RSA TLS Issuing CA 04
   i:C = US, O = DigiCert Inc, OU = www.digicert.com, CN = DigiCert Global Root G2
   a:PKEY: rsaEncryption, 4096 (bit); sigalg: RSA-SHA384
   v:NotBefore: Jun  8 00:00:00 2023 GMT; NotAfter: Aug 25 23:59:59 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIILFjCCCP6gAwIBAgITMwLqY67QAeVfgLCxSQAAAupjrjANBgkqhkiG9w0BAQwF
ADBdMQswCQYDVQQGEwJVUzEeMBwGA1UEChMVTWljcm9zb2Z0IENvcnBvcmF0aW9u
MS4wLAYDVQQDEyVNaWNyb3NvZnQgQXp1cmUgUlNBIFRMUyBJc3N1aW5nIENBIDA0
MB4XDTI1MTAxOTExMjY1N1oXDTI2MDQxNzExMjY1N1owgYYxCzAJBgNVBAYTAlVT
MQswCQYDVQQIEwJXQTEQMA4GA1UEBxMHUmVkbW9uZDEeMBwGA1UEChMVTWljcm9z
b2Z0IENvcnBvcmF0aW9uMTgwNgYDVQQDEy9jcjE1Lm5vcndheWVhc3QxLWEuY29u
dHJvbC5kYXRhYmFzZS53aW5kb3dzLm5ldDCCASIwDQYJKoZIhvcNAQEBBQADggEP
ADCCAQoCggEBAOvQ4UVag/w68SDvMo2yic6v0pLnKejS+h0jSItfspjpr4e0jcqv
6InRUXTo/O3z/5zmbc35eY6CYblNSw/wegC6CEE1hGlNlCiHqv6r00MyetX7rIoG
RCzku8uUMnJkDys/n7/H8vyQ4cryemg+nS0w3b7d8693fLjQQcjPvrsPutQ2f3vV
du2zFpcU660gHL3SZA7M7MNMWNV88Q4nMsdeVjKk07eJVyKyCY+ym09IheV+4lRo
aBQK5SKL+1NjmurvysK4zXzvMC3hYzo/oX8cwo/dKlrdVBh5y739R4M8mAKGIbyR
/2GY04WzOoJfowcNY23NdBT5LEgJBQTI9HUCAwEAAaOCBqMwggafMIIBgAYKKwYB
BAHWeQIEAgSCAXAEggFsAWoAdwCWl2S/VViXrfdDh2g3CEJ36fA61fak8zZuRqQ/
D8qpxgAAAZn8QjPmAAAEAwBIMEYCIQCWij6EtqvQeoEoJTW5Fbs5GTyKN+tNEixA
GZhLlJ/fHQIhAILgxlK3jlG4lE0v1jbQAB9mXYBmdTU6CtTGyGucoZqkAHcAZBHE
bKQS7KeJHKICLgC8q08oB9QeNSer6v7VA8l9zfAAAAGZ/EIzmQAABAMASDBGAiEA
w54PUBAjnaAMFaOl37px4T2Fi7YIouXZg3So8Dtwa78CIQDmUnn5m8bn62HhBgcg
ZlVrbdTFf14Gd+vF9vLbji3K/gB2AA5XlLzzrqk+MxssmQez95Dfm8I9cTIl3SGp
JaxhxU4hAAABmfxCM4oAAAQDAEcwRQIhAKmhm99THQZJNTgZxGNoLKool7uRH6l4
wc280vDtvauuAiAn70lFjqaQ6VSy5apvgEe5DiAUUT4tgSjA6M5w5eOOZDAnBgkr
BgEEAYI3FQoEGjAYMAoGCCsGAQUFBwMCMAoGCCsGAQUFBwMBMDwGCSsGAQQBgjcV
BwQvMC0GJSsGAQQBgjcVCIe91xuB5+tGgoGdLo7QDIfw2h1dgqvnMIft8R8CAWQC
AS0wgbQGCCsGAQUFBwEBBIGnMIGkMHMGCCsGAQUFBzAChmdodHRwOi8vd3d3Lm1p
Y3Jvc29mdC5jb20vcGtpb3BzL2NlcnRzL01pY3Jvc29mdCUyMEF6dXJlJTIwUlNB
JTIwVExTJTIwSXNzdWluZyUyMENBJTIwMDQlMjAtJTIweHNpZ24uY3J0MC0GCCsG
AQUFBzABhiFodHRwOi8vb25lb2NzcC5taWNyb3NvZnQuY29tL29jc3AwHQYDVR0O
BBYEFP1L2Q/CTCz8JZXLJe/hTMEYp8FUMA4GA1UdDwEB/wQEAwIFoDCCAqgGA1Ud
EQSCAp8wggKbgjEqLmNyMTUubm9yd2F5ZWFzdDEtYS5jb250cm9sLmRhdGFiYXNl
LndpbmRvd3MubmV0gipub3J3YXllYXN0MS1hLmNvbnRyb2wuZGF0YWJhc2Uud2lu
ZG93cy5uZXSCLCoubm9yd2F5ZWFzdDEtYS5jb250cm9sLmRhdGFiYXNlLndpbmRv
d3MubmV0ghYqLmRhdGFiYXNlLndpbmRvd3MubmV0gh0qLmRhdGFiYXNlLnNlY3Vy
ZS53aW5kb3dzLm5ldIIgKi5zZWNvbmRhcnkuZGF0YWJhc2Uud2luZG93cy5uZXSC
HCoubWFvaWFkYi5kYXRhYmFzZS5henVvZS5jb22CGioubXlzcWwuZGF0YWJhc2Uu
```

```
YXp1cmUuY29tgh0qLnBvc3RncmVzLmRhdGFiYXNlLmF6dXJlLmNvbYIYKi5zcWwu
cHJvamVjdGFyY2FkaWEubmV0ghYqLnNxbC5henVyZXN5bmFwc2UubmV0gh4qLnNx
bC5henVyZXN5bmFwc2UtZG9nZm9vZC5uZXSCHCouZGF0YWJhc2UtZmxlZXQud2lu
ZG93cy5uZXSCJiouc2Vjb25kYXJ5LmRhdGFiYXNlLWZsZWV0LndpbmRvd3MubmV0
giUqLmRhaWx5LWRhdGFiYXNlLmZhYnJpYy5taWNyb3NvZnQuY29tgiMqLmR4dC1k
YXRhYmFzZS5mYWJyaWMubWljcm9zb2Z0LmNvbYIkKi5tc2l0LWRhdGFiYXNlLmZh
YnJpYy5taWNyb3NvZnQuY29tgh8qLmRhdGFiYXNlLmZhYnJpYy5taWNyb3NvZnQu
Y29tgi9jcjE1Lm5vcndheWVhc3QxLWEuY29udHJvbC5kYXRhYmFzZS53aW5kb3dz
Lm5ldDAMBgNVHRMBAf8EAjAAMGoGA1UdHwRjMGEwX6BdoFuGWWh0dHA6Ly93d3cu
bWljcm9zb2Z0LmNvbS9wa2lvcHMvY3JsL01pY3Jvc29mdCUyMEF6dXJlJTIwUlNB
JTIwVExTJTIwSXNzdWluZyUyMENBJTIwMDQuY3JsMGYGA1UdIARfMF0wUQYMKwYB
BAGCN0yDfQEBMEEwPwYIKwYBBQUHAgEWM2h0dHA6Ly93d3cubWljcm9zb2Z0LmNv
bS9wa2lvcHMvRG9jcy9SZXBvc2l0b3J5Lmh0bTAIBgZngQwBAgIwHwYDVR0jBBgw
FoAUO3DRU+l2JZ1gqMpmD8abrm9UFmowHQYDVR0lBBYwFAYIKwYBBQUHAwIGCCsG
AQUFBwMBMA0GCSqGSIb3DQEBDAUAA4ICAQBJAatCWF/ZqYmllpNDuytr99XYFCWo
mW0+QqdmLw4G3oC+2M+ckHgzPndewUXgN13UvpFzt9wBfJ3IFDOiHUOj8KscnVML
oTe/Q+3USm3eScmifofONs59BCTj8Vrl1uck1lmSbC7rE6iicaI/CBQJ4f5njTGr
m8/mA5U90e64m/qMp1+VyEZrTmuC0ClhWrRHNKepXGN4P0uF+lchJVdJLk7LJTsO
wUgn3y4K03E3g8l+13YhfTtcnROV52z7OewJrLY356GBYJYThBXFnFV8WnogI227
gQ2QA7AVFXD9n5X6CM1kFQIz/XWnWTBr4j/ou2ny00dWLs+GWs9RM5/ecIldi1CI
UOsruJPPJv2Eq0h0t1L8EP/ncnfY2yW/vQR2+h7LX0TUsPXJ1P6weYe8ALr7HtGz
xRI3yhrcRBnWqqbnn93YWRiiuza4aMGSr0OR3tidNCZqK86mxnzdPGO7+FYWuINW
pN9/TN8QgmnkvYfDfly5Z1HQYcnMi9Z8PVczYqz8p/TGjH3MdQeu6Vxz/54cANSM
ZkU1I1Oj9waEWV3Qm1GZW/eB5iRcaFVypd2kB100lUjhabh1liyw7f9qQowsQfBt
umWME2jbuEhJGqCVoiKW7Q7TveckTvvb9yZrO/DjcSSr21i+o60iw5zNYtOak6L5
RsDL9er3nI1Gow==
-----END CERTIFICATE-----
subject=C = US, ST = WA, L = Redmond, O = Microsoft Corporation, CN = cr15.norwayeast1-
a.control.database.windows.net
issuer=C = US, O = Microsoft Corporation, CN = Microsoft Azure RSA TLS Issuing CA 04
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: RSA-PSS
Server Temp Key: ECDH, secp384r1, 384 bits
---
SSL handshake has read 4821 bytes and written 394 bytes
Verification: OK
---
New, TLSv1.2, Cipher is ECDHE-RSA-AES256-GCM-SHA384
Server public key is 2048 bit
Secure Renegotiation IS supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
SSL-Session:
    Protocol  : TLSv1.2
    Cipher    : ECDHE-RSA-AES256-GCM-SHA384
    Session-ID: 813700004EAC879C434CC22029634EF8556217970B5BB2004927CAE199E08C87
    Session-ID-ctx:
    Master-Key:
E09B5F4DA464742BF2342BD785C416D9227E51C8F23B837A16BBB7EC5ED5EC915E381752BA21349898B2FBAE33E925F8
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    Start Time: 1766320270
    Timeout   : 7200 (sec)
    Verify return code: 0 (ok)
    Extended master secret: yes
---
DONE
```

```
azureuser@techstart-vm:~$ curl -I http://131.163.89.9/login
HTTP/1.1 301 Moved Permanently
Date: Wed, 17 Dec 2025 13:32:46 GMT
Server: Apache/2.4.58 (Ubuntu)
Location: https://131.163.89.9/login
Content-Type: text/html; charset=iso-8859-1
```

Figure 42 above: the HTTP response headers to a web request verifies that it is redirected to HTTPS.

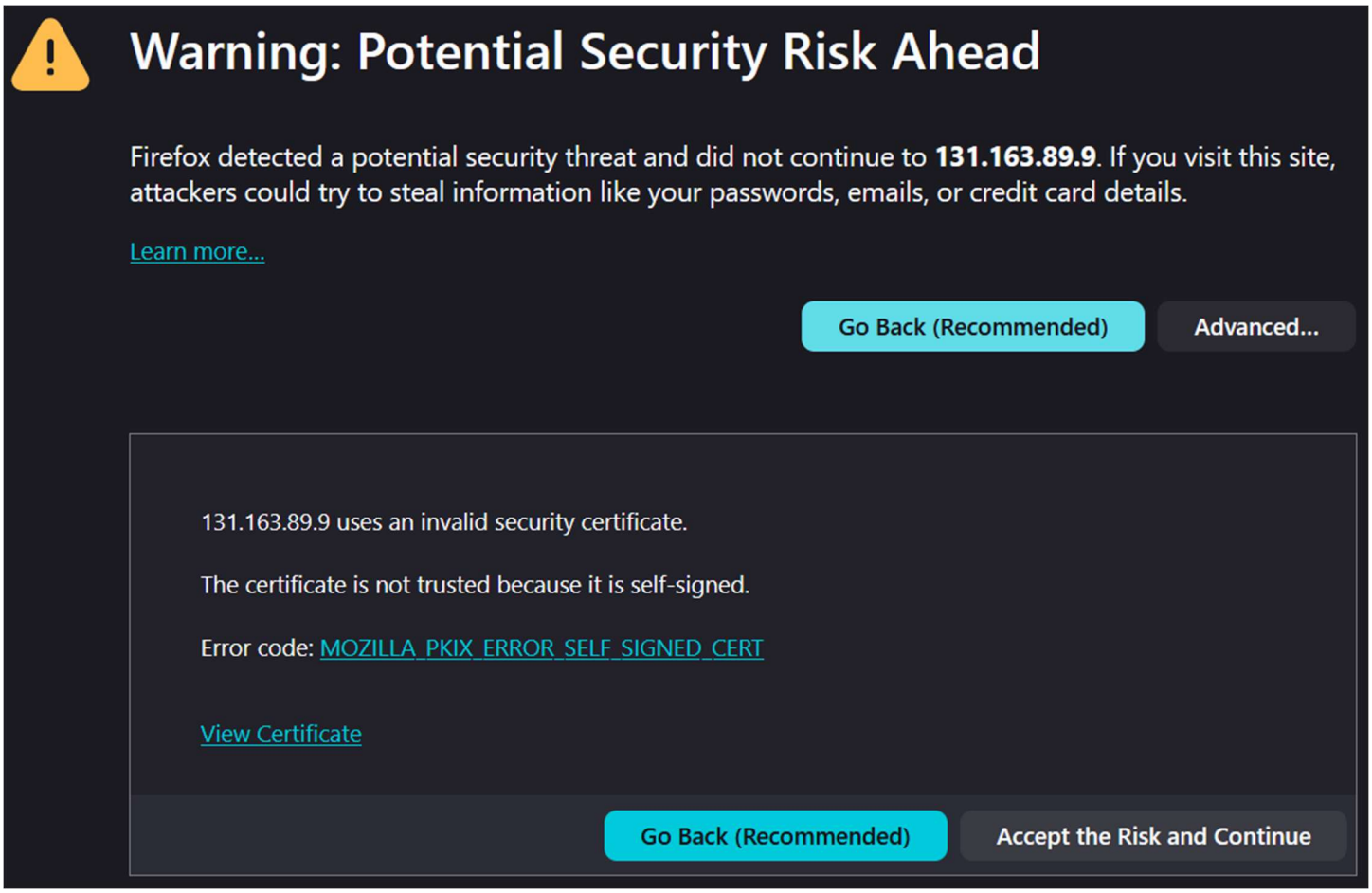


Figure 43 above: browser warning due to self-signed certificate that appears when accessing the webapp site. Requires the user to explicitly accept the risk to continue.

System Hardening and Monitoring

Configuration

Hardening on techstart-vm was implemented using UFW as a simple front-end to Linux's Firewall. UFW was set to a default deny for incoming traffic and a default allow for outgoing traffic. Incoming SSH (22/TCP) traffic was rate limited to protect against brute force attacks by allowing up to five connections attempts within 30 seconds per IP address before denying subsequent connections attempts. Incoming HTTP (80/TCP) and HTTPS (443/TCP) traffic were allowed for the web application. SSH was further protected against brute-force attempts by combining UFW rate limiting on port 22 with Fail2ban running the sshd jail, so repeated authentication failures are automatically detected and the source IP is temporarily banned. The unattended-upgrades package was enabled to automatically apply security updates using apt-daily and apt-daily-upgrade systemd timers, reducing exposure to vulnerabilities over time.

Screenshots

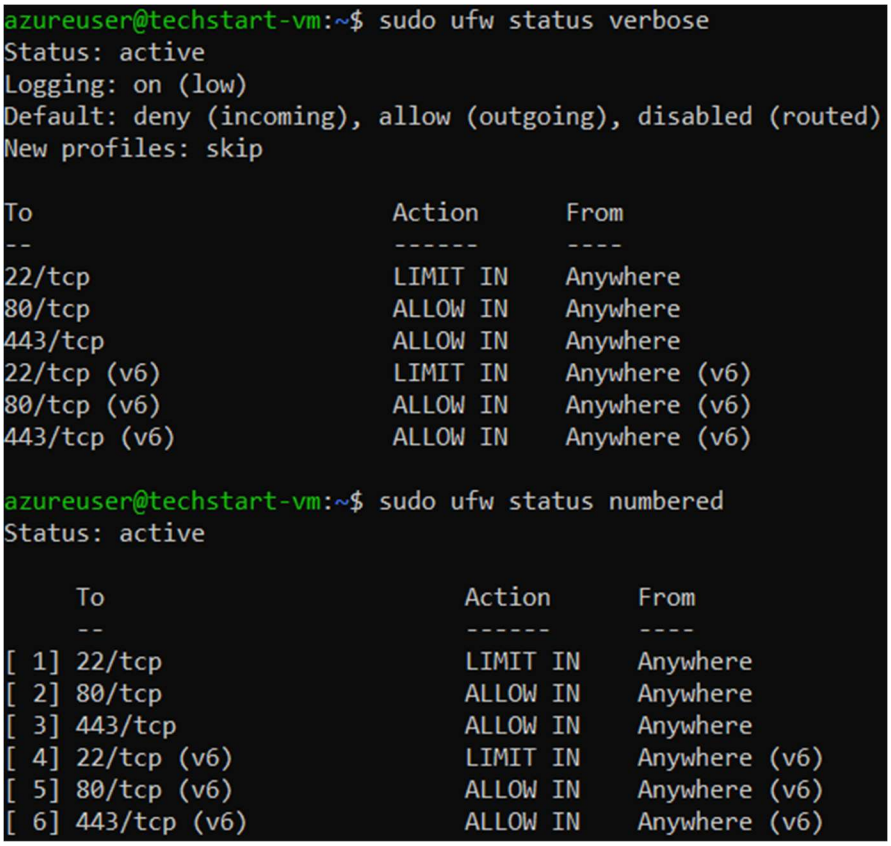


Figure 44 above: UFW firewall status on techstart-vm shows the default policy is to deny incoming traffic and allow outgoing traffic. HTTP (80/tcp) and HTTPS (443/tcp) traffic are explicitly allowed in, while SSH (22/tcp) is rate limited.



Figure 45 above: UFW manual indicates that only up to five SSH connections from the same IP within 30 seconds are allowed in techstart-vm, before denying subsequent connections.

```
azureuser@techstart-vm:~$ sudo systemctl status fail2ban --no-pager
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
     Active: active (running) since Thu 2025-12-18 19:22:18 CET; 2 days ago
       Docs: man:fail2ban(1)
   Main PID: 787 (fail2ban-server)
      Tasks: 5 (limit: 993)
     Memory: 21.4M (peak: 25.7M)
        CPU: 3min 30.868s
     CGroup: /system.slice/fail2ban.service
             └─787 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Dec 18 19:22:18 techstart-vm systemd[1]: Started fail2ban.service - Fail2Ban Service.
Dec 18 19:22:18 techstart-vm fail2ban-server[787]: 2025-12-18 19:22:18,623 fail2ban.configreader   [787]: WARNING 'allowipv6' not defined in 'Definition'. Using default one: 'auto'
Dec 18 19:22:20 techstart-vm fail2ban-server[787]: Server ready
```

Figure 46 above: Fail2Ban service is enabled and running on techstart-vm.

```
azureuser@techstart-vm:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
```

Figure 47 above: sshd jail is active and has not detected suspected brute force attacks.

```
azureuser@techstart-vm:~$ cat /etc/apt/apt.conf.d/20auto-upgrades
APT::Periodic::Update-Package-Lists "1";
APT::Periodic::Unattended-Upgrade "1";
```

Figure 48 above: periodic security updates and unattended upgrades are enabled.

```
azureuser@techstart-vm:~$ systemctl is-enabled apt-daily.timer apt-daily-upgrade.timer
enabled
enabled
```

Figure 49 above: timers to check updates and apply the upgrades are enabled.

```
azureuser@techstart-vm:~$ systemctl list-timers | grep -E "apt-daily|apt-daily-upgrade"
Thu 2025-12-18 23:52:36 CET     4h 37min Thu 2025-12-18 11:23:33 CET       7h ago apt-daily.timer              apt-daily.service
Fri 2025-12-19 06:13:57 CET          10h Thu 2025-12-18 06:59:33 CET      12h ago apt-daily-upgrade.timer      apt-daily-upgrade.service
```

Figure 50 above: both timers are scheduled to run at least once daily.

Verification

```
PS C:\Users\35383> Test-NetConnection 131.163.89.9 -Port 8080
WARNING: TCP connect to (131.163.89.9 : 8080) failed
WARNING: Ping to 131.163.89.9 failed with status: TimedOut


ComputerName            : 131.163.89.9
RemoteAddress           : 131.163.89.9
RemotePort              : 8080
InterfaceAlias          : WiFi
SourceAddress           : 192.168.1.7
PingSucceeded           : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded        : False
```

Figure 51 above: connection to techstart-vm is blocked on ports not allowed in UFW incoming rules.

```
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
azureuser@131.163.89.9: Permission denied (publickey).
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
azureuser@131.163.89.9: Permission denied (publickey).
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
azureuser@131.163.89.9: Permission denied (publickey).
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
azureuser@131.163.89.9: Permission denied (publickey).
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
azureuser@131.163.89.9: Permission denied (publickey).
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
ssh: connect to host 131.163.89.9 port 22: Connection timed out
```

Figure 52 above: five SSH connections each leading to failed authentication attempts before being blocked.

```
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -i techstart-vm_key.pem -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Dec 17 19:20:32 CET 2025

  System load:  0.0               Processes:             124
  Usage of /:   17.0% of 28.02GB  Users logged in:       1
  Memory usage: 52%               IPv4 address for eth0: 10.0.1.4
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

27 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


Last login: Wed Dec 17 19:20:28 2025 from [redacted]
azureuser@techstart-vm:~$ exit
logout
Connection to 131.163.89.9 closed.
PS C:\Users\35383\Documents\0Uni\Semester 3\Security Administration\Project> ssh -i techstart-vm_key.pem -L 5000:127.0.0.1:5000 azureuser@131.163.89.9
ssh: connect to host 131.163.89.9 port 22: Connection timed out
```

Figure 53 above: successful authentication attempts also get blocked after the fifth SSH connection within 30 seconds.

```
azureuser@techstart-vm:~$ sudo tail -n 50 /var/log/fail2ban.log
2025-12-18 01:20:10,222 fail2ban.server         [52535]: INFO    --------------------------------------------------
2025-12-18 01:20:10,224 fail2ban.server         [52535]: INFO    Starting Fail2ban v1.0.2
2025-12-18 01:20:10,227 fail2ban.observer       [52535]: INFO    Observer start...
2025-12-18 01:20:10,240 fail2ban.database       [52535]: INFO    Connected to fail2ban persistent database '/var/lib/fail2ban/fail2ban.sqlite3'
2025-12-18 01:20:10,253 fail2ban.database       [52535]: WARNING New database created. Version '4'
2025-12-18 01:20:10,256 fail2ban.jail           [52535]: INFO    Creating new jail 'sshd'
2025-12-18 01:20:10,657 fail2ban.jail           [52535]: INFO    Jail 'sshd' uses systemd {}
2025-12-18 01:20:10,659 fail2ban.jail           [52535]: INFO    Initiated 'systemd' backend
2025-12-18 01:20:10,662 fail2ban.filter         [52535]: INFO      maxLines: 1
2025-12-18 01:20:10,699 fail2ban.filtersystemd  [52535]: INFO    [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.service + _COMM=sshd'
2025-12-18 01:20:10,700 fail2ban.filter         [52535]: INFO      maxRetry: 5
2025-12-18 01:20:10,700 fail2ban.filter         [52535]: INFO      findtime: 600
2025-12-18 01:20:10,700 fail2ban.actions        [52535]: INFO      banTime: 600
2025-12-18 01:20:10,700 fail2ban.filter         [52535]: INFO      encoding: UTF-8
2025-12-18 01:20:10,703 fail2ban.jail           [52535]: INFO    Jail 'sshd' started
2025-12-18 01:20:10,706 fail2ban.filtersystemd  [52535]: INFO    [sshd] Jail is in operation now (process new journal entries)
```

Figure 54 above: Fail2ban logs verify the sshd jail is operating.

```
azureuser@techstart-vm:~$ sudo tail -n 80 /var/log/unattended-upgrades/unattended-upgrades.log
2025-12-12 06:36:34,192 INFO Starting unattended upgrades script
2025-12-12 06:36:34,192 INFO Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security
2025-12-12 06:36:34,192 INFO Initial blacklist:
2025-12-12 06:36:34,192 INFO Initial whitelist (not strict):
2025-12-12 06:36:36,839 INFO Packages that will be upgraded: python3-urllib3
2025-12-12 06:36:36,839 INFO Writing dpkg log to /var/log/unattended-upgrades/unattended-upgrades-dpkg.log
2025-12-12 06:36:43,382 INFO All upgrades installed
2025-12-12 07:54:49,819 INFO Starting unattended upgrades script
2025-12-12 07:54:49,821 INFO Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security
2025-12-12 07:54:49,821 INFO Initial blacklist:
2025-12-12 07:54:49,821 INFO Initial whitelist (not strict):
2025-12-13 06:05:34,010 INFO Starting unattended upgrades script
2025-12-13 06:05:34,011 INFO Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security
2025-12-13 06:05:34,011 INFO Initial blacklist:
2025-12-13 06:05:34,011 INFO Initial whitelist (not strict):
2025-12-13 06:05:35,258 INFO No packages found that can be upgraded unattended and no pending auto-removals
2025-12-13 16:57:54,140 INFO Starting unattended upgrades script
2025-12-13 16:57:54,141 INFO Allowed origins are: o=Ubuntu,a=noble, o=Ubuntu,a=noble-security, o=UbuntuESMApps,a=noble-apps-security, o=UbuntuESM,a=noble-infra-security
```

Figure 55 above: logs show the unattended-upgrades package is installing updates.

Bash Scripting

A security monitoring bash script was added to monitor the system for recent suspicious activities such as failed SSH attempts, UFW blocks, Fail2ban bans and sudo commands. The script outputs basic host context and then validates that key security services (sshd, ufw, fail2ban, cron, apache2) are running. It also checks permissions on critical files such as /etc/shadow and /etc/ssh/sshd_config as well as home directories for overly permissive access. This provides a lightweight and repeatable health and security snapshot that can be run on demand or scheduled using cron for ongoing monitoring.

```
 GNU nano 7.2                                                    /usr/local/sbin/security_monitor.sh
#!/bin/bash

echo "=== TechStart Solutions Security Monitor ==="
echo "Current username: $(whoami)"
echo "User ID: $(id -u)"
echo "Hostname: $(hostname)"
echo "Current date and time: $(date)"
echo "Number of logged-in users: $(who | wc -l)"
echo "Uptime: $(uptime)"
echo ""

check_service () {
    SERVICE_NAME=$1

    if systemctl is-active --quiet "$SERVICE_NAME"; then
        echo "$SERVICE_NAME is running"
    else
        echo "$SERVICE_NAME is NOT running"
        return 1
    fi
}

echo "=== Service checks ==="
check_service "ssh"
check_service "ufw"
check_service "cron"
check_service "fail2ban"
check_service "apache2"
echo""

check_file_permissions () {
    FILENAME=$1
    perms=$(stat -c %a "$FILENAME")
    echo "File: $FILENAME | Permissions: $perms"

    if [ "$FILENAME" == "/etc/shadow" ] && { [ "$perms" -ne 640 ] && [ "$perms" -ne 600 ]; }; then #-ne = not equal to
        echo "Shadow file may be insecure"
        return 1
    fi

    if [ "$FILENAME" != "/etc/shadow" ] && [ "$perms" -gt 644 ]; then #-gt = greater than
        echo "$FILENAME permissions may be too open"
        return 1
    fi
}

echo "=== Critical file permission checks ==="
for FILENAME in /etc/passwd /etc/shadow /etc/ssh/sshd_config; do
    check_file_permissions "$FILENAME"
done
echo ""

check_home_permissions () {
    user=$1
    if [ -d "/home/$user" ]; then
        echo "$user: Home directory exists"
        perms=$(stat -c %a "/home/$user")
        if [ "$perms" -gt 750 ]; then
            echo "$user's home directory has permissions: $perms"
            return 1
        fi
    else
        echo "$user: Home directory MISSING"
        return 1
    fi
}

echo "=== Home directory permission checks ==="
for user in azureuser dev1 dev2 qa1 qa2 pm1 pm2; do
    check_home_permissions "$user"
done
echo ""

echo "=== Suspicious activity within the last 30 minutes ==="
#checks failled ssh  attempts
FAILED_SSH=$(journalctl -u sshd --since "30 min ago" --no-pager 2>/dev/null | \
  grep -Eic "Failed password|Invalid user|authentication failure|Disconnected from authenticating user")
echo "Failed/invalid SSH attempts: $FAILED_SSH"

#checks fail2ban bans
if command -v fail2ban-client >/dev/null 2>&1; then
  BANNED=$(fail2ban-client status sshd 2>/dev/null | awk -F': ' '/Currently banned/ {print $2}')
  echo "Fail2ban currently banned IPs: ${BANNED:-0}"
fi

#checks blocks by ufw
UFW_BLOCKS=$(journalctl --since "30 min ago" --no-pager 2>/dev/null | grep -c "UFW BLOCK")
echo "UFW BLOCK events: $UFW_BLOCKS"

#checks sudo commands withtin last 30 mins
SUDO_EVENTS=$(journalctl -t sudo --since "30 min ago" --no-pager 2>/dev/null | wc -l)
echo "sudo events: $SUDO_EVENTS"
```

Figure 56 above: The full source code in the security monitoring bash script in techstart-vm.

```
azureuser@techstart-vm:~$ sudo /usr/local/sbin/security_monitor.sh
=== TechStart Solutions Security Monitor ===
Current username: root
User ID: 0
Hostname: techstart-vm
Current date and time: Wed Dec 17 22:11:31 CET 2025
Number of logged-in users: 2
Uptime:  22:11:31 up 12 days,  3:51,  1 user,  load average: 0.00, 0.00, 0.00

=== Service checks ===
ssh is running
ufw is running
cron is running
fail2ban is running
apache2 is running

=== Critical file permission checks ===
File: /etc/passwd | Permissions: 644
File: /etc/shadow | Permissions: 640
File: /etc/ssh/sshd_config | Permissions: 644

=== Home directory permission checks ===
azureuser: Home directory exists
dev1: Home directory exists
dev2: Home directory exists
qa1: Home directory exists
qa2: Home directory exists
pm1: Home directory exists
pm2: Home directory exists

=== Suspicious activity within the last 30 minutes ===
Failed/invalid SSH attempts: 0
Fail2ban currently banned IPs: 0
UFW BLOCK events: 0
sudo events: 1718
```

Figure 57 above: security_monitor.sh is ran on techstart-vm and outputs the basic host context, confirms all security services are running and none of the critical files nor home directories are overly permissive (greater than 644), and reports that in the past 30 minutes there have been 1718 sudo-related log lines that could indicate automation or possible abuse, but no banned IPs, failed SSH attempts, nor UFW blocks.

Logging, Backup, and Audit Trail

Configuration

System logging on techstart-vm was configured to create a reliable audit trail for authentication events and service activity using both systemd-journald for structured binary logs with detailed metadata that makes querying more powerful, and rsyslog for plain text logs with simpler metadata. The necessary rsyslog configurations were already set and could be seen by entering commands to verify it is enabled and keeping logs in traditional files such as /var/log/syslog and /var/log/auth.log to help review tools such as Fail2ban and logrotate. journald was configured for the logs to start deleting if it reaches 200MB or if the free space drops to 200MB. If the storage and vacancy capacity has not been reached, the logs will be stored for 30 days even while power is off as persistent storage was configured.

The automated backups for TechStartDB were already configured which could be seen on techstartsolutions SQL server's Backup settings which by default performs automated backups (including differential backups every 24 hours) and allows restoration of the database for up to seven days in cases of accidental or malicious data loss. No addition configurations were made as a longer point-in-time restore would be more costly, increasing retention would use more storage, and turning on time-based immutability would force us to wait a certain period of time before being able to delete a backup with misconfigurations.

Backup policies for techstart-vm were implemented by scheduling a cron job to regularly backup the web application and key configuration files such as the Apache configuration and the TLS certificate. The scheduled backup runs automatically using cron and writes timestamped and compressed archives into a backup directory so the web server can be restored after deletion or misconfiguration.

Log retention on techstart-vm was implemented using logrotate to prevent /var/log/ from growing indefinitely while keeping enough history for troubleshooting and auditing. The global logrotate policy in /etc/logrotate.conf is enabled by using the system logrotate timer and setting a baseline of weekly rotation with four weeks archives, creating new log files after every rotation. Service-specific retention is in the folder /etc/logrotate.d/ where most core security logs such as rsyslog and UFW were already preconfigured to rotate and compress logs every week. Only changes were made in /etc/logrotate.d/apache2 where Apache access and error logs were configured to rotate daily and to keep 30 rotations.

Screenshots

```
  GNU nano 7.2                                                                /etc/rsyslog.conf
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf
#################
#### MODULES ####
#################

module(load="imuxsock") # provides support for local system logging
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
#module(load="imudp")
#input(type="imudp" port="514")

# provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")

###########################
#### GLOBAL DIRECTIVES ####
###########################

# Filter duplicated messages
$RepeatedMsgReduction on
#
# Set the default permissions for all log files.
#
$FileOwner syslog
$FileGroup adm
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022
$PrivDropToUser syslog
$PrivDropToGroup syslog
#
# Where to place spool and state files
#
$WorkDirectory /var/spool/rsyslog
#
# Include all config files in /etc/rsyslog.d/
#
$IncludeConfig /etc/rsyslog.d/*.conf
```

Figure 58 above: rsyslog is already configured in the file /etc/rsyslog.conf where no changes were made.

```
  GNU nano 7.2                                                    /etc/systemd/journald.conf
#  This file is part of systemd.
#
#  systemd is free software; you can redistribute it and/or modify it under the
#  terms of the GNU Lesser General Public License as published by the Free
#  Software Foundation; either version 2.1 of the License, or (at your option)
#  any later version.
#
# Entries in this file show the compile time defaults. Local configuration
# should be created by either modifying this file (or a copy of it placed in
# /etc/ if the original file is shipped in /usr/), or by creating "drop-ins" in
# the /etc/systemd/journald.conf.d/ directory. The latter is generally
# recommended. Defaults can be restored by simply deleting the main
# configuration file and all drop-ins located in /etc/.
#
# Use 'systemd-analyze cat-config systemd/journald.conf' to display the full config.
#
# See journald.conf(5) for details.

[Journal]
#to keep data even after power off:
Storage=persistent
#Compress=yes
#Seal=yes
#SplitMode=uid
#SyncIntervalSec=5m
#RateLimitIntervalSec=30s
#RateLimitBurst=10000
#so logs don't fill the disk:
SystemMaxUse=200M
#journald will stop growing earlier if free space drops to 200MB:
SystemKeepFree=200M
#SystemMaxFileSize=
#SystemMaxFiles=100
#RuntimeMaxUse=
#RuntimeKeepFree=
#RuntimeMaxFileSize=
#RuntimeMaxFiles=100
#keeps logs up to 30 days if it does not reach the storage capacity before:
MaxRetentionSec=30day
```

Figure 59 above: journald is configured in the file /etc/systemd/journald.conf to enable persistent storage, limit the storage the log uses and the retention.

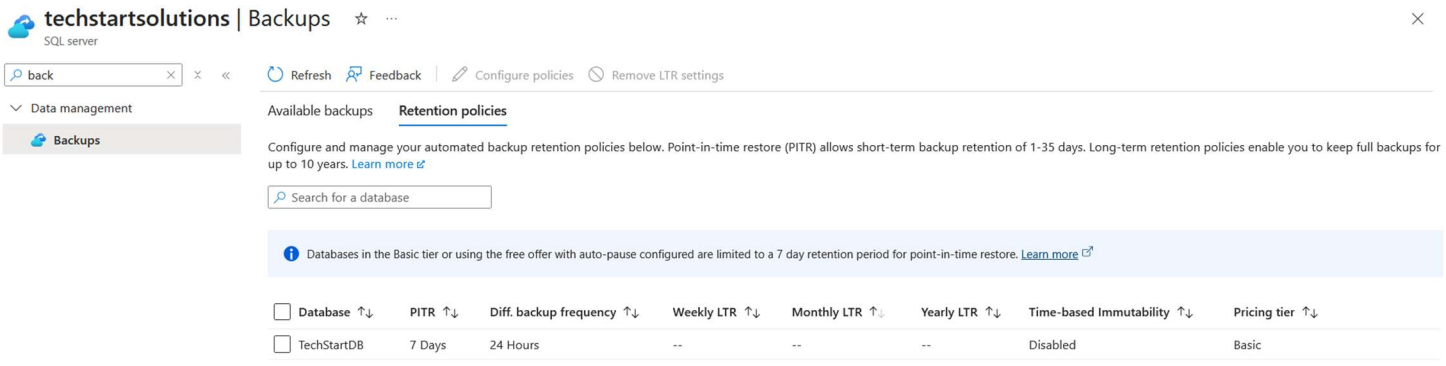


Figure 60 above: automated backups for TechStartDB were already configured and set to when the SQL server and database was created.

```
  GNU nano 7.2                                                    /usr/local/bin/backup_webapp.sh
#!/bin/bash
set -e
DEST="/var/backups/webapp"
mkdir -p "$DEST"
tar -czf "$DEST/webapp-$(date +%F-%H%M).tar.gz" /home/azureuser/webapp
```

Figure 61 above: bash script that creates backup folder and timestamped compressed files of daily archives of the web application.

```
azureuser@techstart-vm:~$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/backup_webapp.sh >> /var/log/backup_webapp.log 2>&1
```

Figure 62 above: cron job that runs the bash script daily at 2am and appends the output (including errors) to the log file.

```
  GNU nano 7.2                                                    /etc/logrotate.conf
# see "man logrotate" for details

# global options do not affect preceding include directives

# rotate log files weekly
weekly

# use the adm group by default, since this is the owning group
# of /var/log/.
su root adm

# keep 4 weeks worth of backlogs
rotate 4

# create new (empty) log files after rotating old ones
create

# use date as a suffix of the rotated file
#dateext

# uncomment this if you want your log files compressed
#compress

# packages drop log rotation information into this directory
include /etc/logrotate.d

# system-specific logs may also be configured here.
```

Figure 63 above: the global logrotate policy is configured to rotate logs weekly, keep 4 rotations/weeks of logs, and create new log files after rotation.

```
  GNU nano 7.2                                                /etc/logrotate.d/apache2
/var/log/apache2/*.log {
        daily
        missingok
        rotate 30
        compress
        delaycompress
        notifempty
        create 640 root adm
        sharedscripts
        prerotate
                if [ -d /etc/logrotate.d/httpd-prerotate ]; then
                        run-parts /etc/logrotate.d/httpd-prerotate
                fi
        endscript
        postrotate
                if pgrep -f ^/usr/sbin/apache2 > /dev/null; then
                        invoke-rc.d apache2 reload 2>&1 | logger -t apache2.logrotate
                fi
        endscript
}
```

Figure 64 above: the Apache service logrotate policy is configured to rotate logs daily, keep 30 rotations/days of logs, compress rotated log files, and reload Apache after rotation.

Verification

```
azureuser@techstart-vm:~$ systemctl status rsyslog --no-pager
● rsyslog.service - System Logging Service
     Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
     Active: active (running) since Thu 2025-12-18 19:22:20 CET; 42min ago
TriggeredBy: ● syslog.socket
       Docs: man:rsyslogd(8)
             man:rsyslog.conf(5)
             https://www.rsyslog.com/doc/
    Process: 794 ExecStartPre=/usr/lib/rsyslog/reload-apparmor-profile (code=exited, status=0/SUCCESS)
   Main PID: 823 (rsyslogd)
      Tasks: 4 (limit: 993)
     Memory: 3.2M (peak: 5.1M)
        CPU: 165ms
     CGroup: /system.slice/rsyslog.service
             └─823 /usr/sbin/rsyslogd -n -iNONE

Dec 18 19:22:18 techstart-vm systemd[1]: Starting rsyslog.service - System Logging Service...
Dec 18 19:22:20 techstart-vm rsyslogd[823]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.  [v8.2312.0]
Dec 18 19:22:20 techstart-vm rsyslogd[823]: rsyslogd's groupid changed to 102
Dec 18 19:22:20 techstart-vm rsyslogd[823]: rsyslogd's userid changed to 102
Dec 18 19:22:20 techstart-vm rsyslogd[823]: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="823" x-info="https://www.rsyslog.com"] start
Dec 18 19:22:20 techstart-vm systemd[1]: Started rsyslog.service - System Logging Service.
```

Figure 65 above: rsyslog system logging service is enabled and running on techstart-vm.

```
azureuser@techstart-vm:~$ sudo tail -n 20 /var/log/auth.log
2025-12-18T19:45:22.683278+01:00 techstart-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
2025-12-18T19:50:17.730329+01:00 techstart-vm sudo: pam_unix(sudo:session): session closed for user root
2025-12-18T19:55:01.974385+01:00 techstart-vm CRON[1488]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2025-12-18T19:55:01.978071+01:00 techstart-vm CRON[1488]: pam_unix(cron:session): session closed for user root
2025-12-18T19:58:34.967622+01:00 techstart-vm sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/bin/nano /etc/logrotate.d/apache2
2025-12-18T19:58:34.975273+01:00 techstart-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
2025-12-18T19:59:55.511459+01:00 techstart-vm sudo: pam_unix(sudo:session): session closed for user root
2025-12-18T20:02:13.540577+01:00 techstart-vm sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/sbin/logrotate -d /etc/logrotate.conf
2025-12-18T20:02:13.541994+01:00 techstart-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
2025-12-18T20:02:13.572721+01:00 techstart-vm sudo: pam_unix(sudo:session): session closed for user root
2025-12-18T20:03:18.423224+01:00 techstart-vm sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/sbin/logrotate -f /etc/logrotate.d/apache2
2025-12-18T20:03:18.424967+01:00 techstart-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
2025-12-18T20:03:18.548055+01:00 techstart-vm sudo: pam_unix(sudo:session): session closed for user root
2025-12-18T20:03:39.128437+01:00 techstart-vm sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/bin/tail -n 30 /var/lib/logrotate/status
2025-12-18T20:03:39.133227+01:00 techstart-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
2025-12-18T20:03:39.135262+01:00 techstart-vm sudo: pam_unix(sudo:session): session closed for user root
2025-12-18T20:05:01.984834+01:00 techstart-vm CRON[1619]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2025-12-18T20:05:01.991094+01:00 techstart-vm CRON[1619]: pam_unix(cron:session): session closed for user root
2025-12-18T20:05:04.895738+01:00 techstart-vm sudo: azureuser : TTY=pts/0 ; PWD=/home/azureuser ; USER=root ; COMMAND=/usr/bin/tail -n 20 /var/log/auth.log
2025-12-18T20:05:04.896867+01:00 techstart-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by azureuser(uid=1000)
```

Figure 66 above: logs are being written to traditional text log files.

```
azureuser@techstart-vm:~$ systemctl status systemd-journald --no-pager
● systemd-journald.service - Journal Service
     Loaded: loaded (/usr/lib/systemd/system/systemd-journald.service; static)
    Drop-In: /usr/lib/systemd/system/systemd-journald.service.d
             └─nice.conf
     Active: active (running) since Thu 2025-12-18 19:44:05 CET; 20min ago
TriggeredBy: ● systemd-journald.socket
             ● systemd-journald-dev-log.socket
             o systemd-journald-audit.socket
       Docs: man:systemd-journald.service(8)
             man:journald.conf(5)
   Main PID: 1447 (systemd-journal)
     Status: "Processing requests..."
      Tasks: 1 (limit: 993)
   FD Store: 17 (limit: 4224)
     Memory: 1.4M (peak: 1.6M)
        CPU: 39ms
     CGroup: /system.slice/systemd-journald.service
             └─1447 /usr/lib/systemd/systemd-journald

Dec 18 19:44:05 techstart-vm systemd-journald[1447]: Collecting audit messages is disabled.
Dec 18 19:44:05 techstart-vm systemd-journald[1447]: Journal started
Dec 18 19:44:05 techstart-vm systemd-journald[1447]: System Journal (/var/log/journal/c6a770c9b7394af3a7c209fa9ca83c98) is 31.1M, max 200.0M, 168.8M free.
```

Figure 67 above: systemd-journald logging service is enabled and running on techstart-vm.

```
azureuser@techstart-vm:~$ sudo journalctl --disk-usage
Archived and active journals take up 31.1M in the file system.
```

Figure 68 above: the systemd journal logs used 31.1MB of the disk.

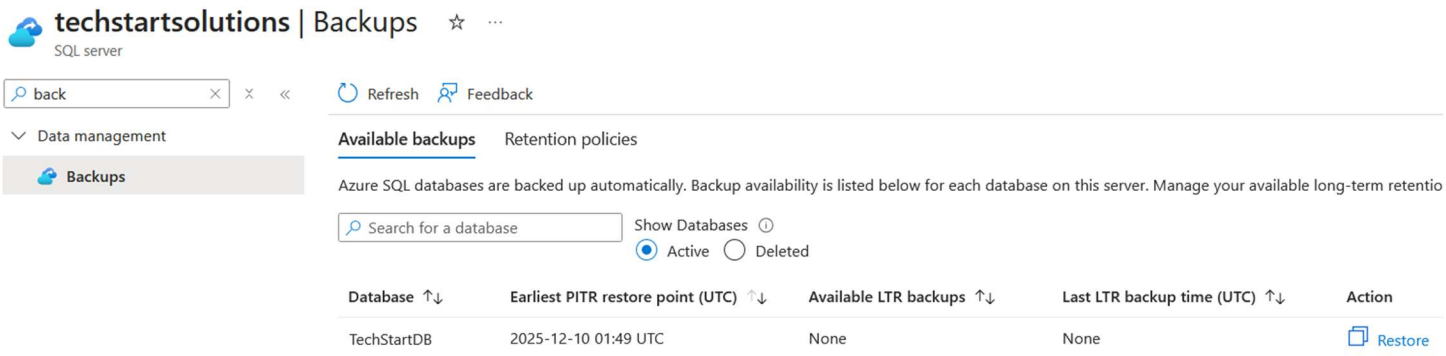


Figure 69 above: TechStartDB's earliest available backup was a week before screenshot was taken.

```
azureuser@techstart-vm:/$ sudo ls -lh /var/backups/webapp/
total 47M
-rw-r--r-- 1 root root 5.9M Dec 11 02:00 webapp-2025-12-11-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 12 02:00 webapp-2025-12-12-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 13 02:00 webapp-2025-12-13-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 14 02:00 webapp-2025-12-14-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 15 02:00 webapp-2025-12-15-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 16 02:00 webapp-2025-12-16-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 17 02:00 webapp-2025-12-17-0200.tar.gz
-rw-r--r-- 1 root root 5.9M Dec 18 02:00 webapp-2025-12-18-0200.tar.gz
```

Figure 70 above: timestamped compressed archive files of the web application are successfully saved in the backup folder where only the owning root user can write.

```
azureuser@techstart-vm:~$ ls -lh /var/log/apache2/
total 200K
-rw-r----- 1 root adm     0 Dec 13 01:00 access.log
-rw-r----- 1 root adm   963 Dec 12 01:01 access.log.1
-rw-r----- 1 root adm   319 Dec 18 20:05 error.log
-rw-r----- 1 root adm   453 Dec 18 20:05 error.log.1
-rw-r----- 1 root adm   555 Dec 18 20:03 error.log.2.gz
-rw-r----- 1 root adm   286 Dec 18 01:00 error.log.3.gz
-rw-r----- 1 root adm   286 Dec 17 01:00 error.log.4.gz
-rw-r----- 1 root adm   284 Dec 16 01:00 error.log.5.gz
-rw-r----- 1 root adm   284 Dec 15 01:00 error.log.6.gz
-rw-r----- 1 root adm   284 Dec 14 01:00 error.log.7.gz
-rw-r----- 1 root adm   25K Dec 13 01:00 error.log.8.gz
-rw-r----- 1 root adm     0 Dec 18 20:03 flask-access.log
-rw-r----- 1 root adm   25K Dec 18 19:28 flask-access.log.1
-rw-r--r-- 1 root root  237 Dec 12 01:32 flask-access.log.2.gz
-rw-r----- 1 root adm     0 Dec 18 20:03 flask-error.log
-rw-r----- 1 root adm   667 Dec 18 02:50 flask-error.log.1
-rw-r--r-- 1 root root  299 Dec 12 01:15 flask-error.log.2.gz
-rw-r----- 1 root adm     0 Dec 18 20:05 flask-ssl-access.log
-rw-r----- 1 root adm   147 Dec 18 20:03 flask-ssl-access.log.1
-rw-r----- 1 root adm  2.3K Dec 18 19:56 flask-ssl-access.log.2.gz
-rw-r--r-- 1 root root  381 Dec 12 01:33 flask-ssl-access.log.3.gz
-rw-r----- 1 root adm   351 Dec 18 20:05 flask-ssl-error.log
-rw-r----- 1 root adm   351 Dec 18 20:03 flask-ssl-error.log.1
-rw-r----- 1 root adm   399 Dec 18 19:22 flask-ssl-error.log.2.gz
-rw-r----- 1 root adm   219 Dec 17 01:00 flask-ssl-error.log.3.gz
-rw-r----- 1 root adm   217 Dec 16 01:00 flask-ssl-error.log.4.gz
-rw-r----- 1 root adm   218 Dec 15 01:00 flask-ssl-error.log.5.gz
-rw-r----- 1 root adm   218 Dec 14 01:00 flask-ssl-error.log.6.gz
-rw-r--r-- 1 root root 1.1K Dec 13 01:00 flask-ssl-error.log.7.gz
-rw-r----- 1 root adm     0 Dec 13 01:00 other_vhosts_access.log
-rw-r----- 1 root adm   115 Dec 12 00:21 other_vhosts_access.log.1
-rw-r----- 1 root adm     0 Dec 13 01:00 ssl_access.log
-rw-r--r-- 1 root root 1.1K Dec 12 01:01 ssl_access.log.1
-rw-r----- 1 root adm   351 Dec 18 20:05 ssl_error.log
-rw-r----- 1 root adm   351 Dec 18 20:03 ssl_error.log.1
-rw-r----- 1 root adm   316 Dec 18 19:22 ssl_error.log.2.gz
-rw-r----- 1 root adm   219 Dec 17 01:00 ssl_error.log.3.gz
-rw-r----- 1 root adm   218 Dec 16 01:00 ssl_error.log.4.gz
-rw-r----- 1 root adm   219 Dec 15 01:00 ssl_error.log.5.gz
-rw-r----- 1 root adm   218 Dec 14 01:00 ssl_error.log.6.gz
-rw-r----- 1 root adm   217 Dec 13 01:00 ssl_error.log.7.gz
-rw-r--r-- 1 root root  741 Dec 12 01:31 ssl_error.log.8.gz
-rw-r----- 1 root adm     0 Dec 13 01:00 techstart-access.log
-rw-r--r-- 1 root root  600 Dec 12 00:13 techstart-access.log.1
-rw-r----- 1 root adm     0 Dec 13 01:00 techstart-error.log
-rw-r--r-- 1 root root 1.4K Dec 12 00:13 techstart-error.log.1
```

Figure 71 above: Apache access and error logs are rotated and compressed into numbered archives.

Access Control Matrix

Team	Azure Role	OS Access	Database Access	Blob Storage Access
Developers	Contributor	sudo group	db_owner	Read/Write
QA Testers	Reader	qa group (no sudo)	db_datareader	Read
Project Managers	Reader	pms group (no sudo)	db_datareader	None

Additional Security Measures

Azure Security Center

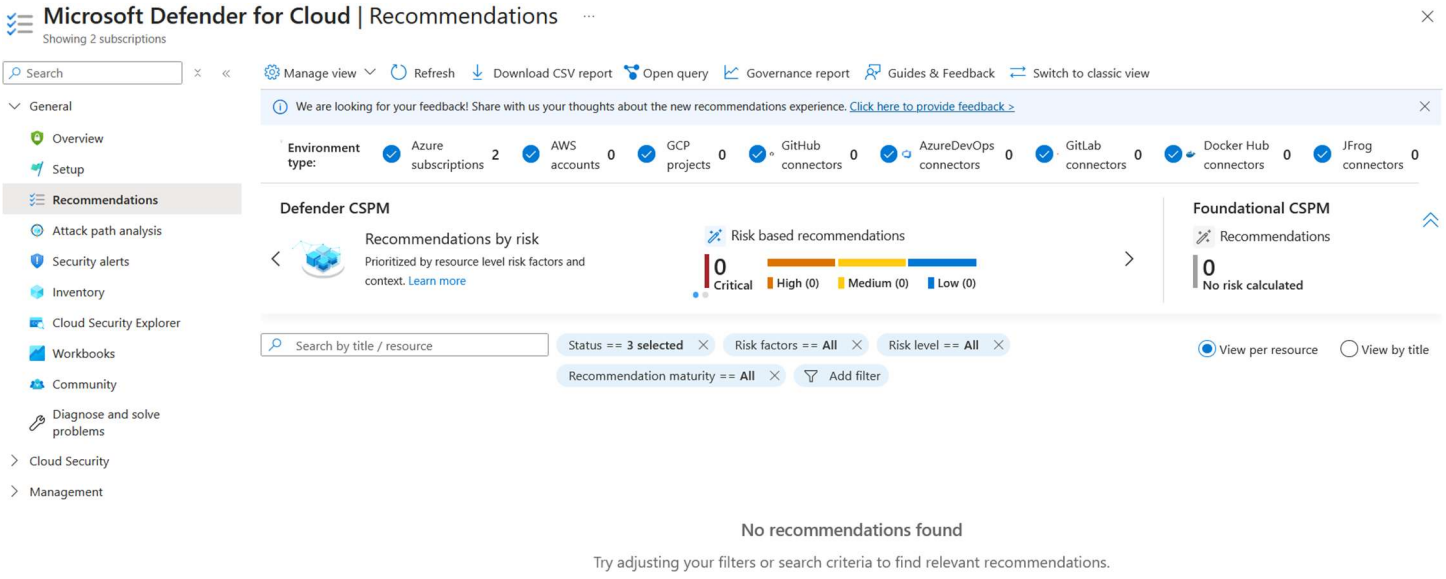
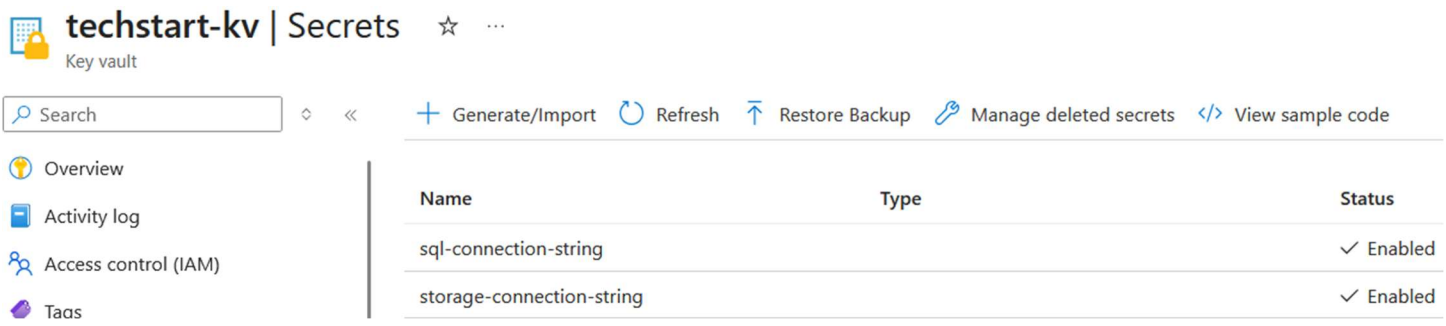
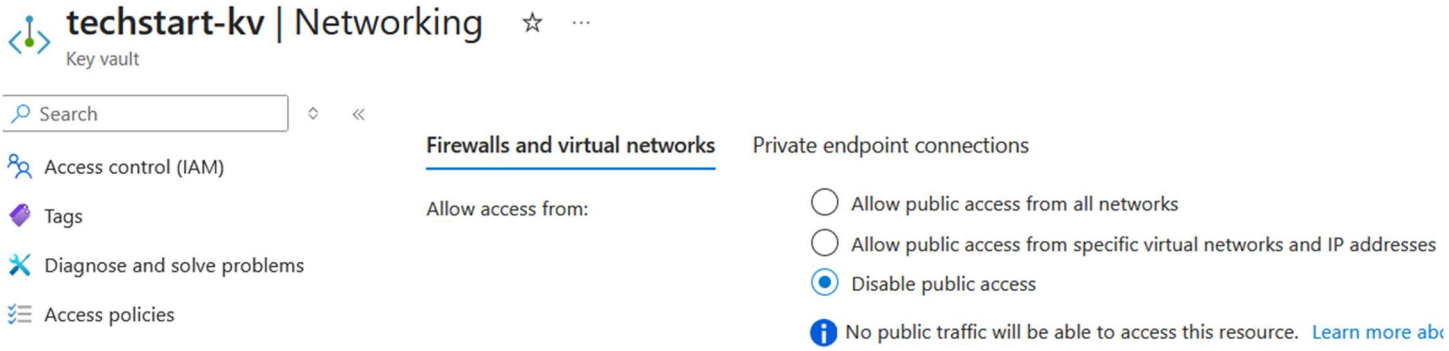


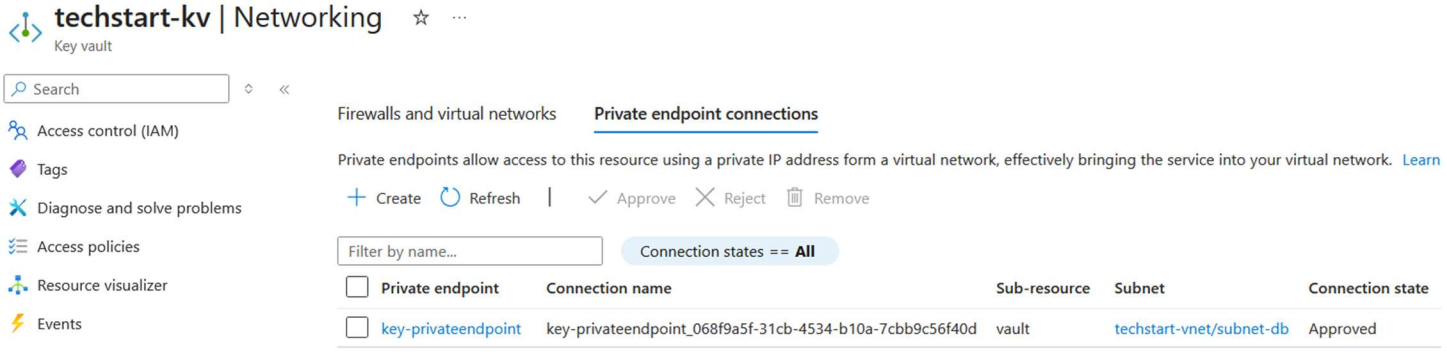
Figure 72 above: no incomplete security recommendations in Microsoft Defender for Cloud.

Azure Key Vault

Key Vault (techstart-kv) was created to store secrets consisting of the SQL connection string and storage connection string, instead of hard-coding sensitive data in files on techstart-vm. Access to secret values was controlled using Azure RBAC so that only developers assigned the Key Vault Secrets User role could read them to follow the principle of Least Privilege. The vault was configured to disable public access and to use a private endpoint connection (key-privateendpoint) which has a network interface with the IP address 10.0.2.5.




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Allow public access from specific virtual networks and IP addresses
Disable public access
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Refresh
Approve
Reject
Remove
Filter by name...
Connection states == All
Private endpoint
Connection name
Sub-resource
Subnet
Connection state
key-privateendpoint
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vault
techstart-vnet/subnet-db
Approved